THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS
AZOR - ISRAEL

GENERAL ASPECTS AND INITIAL RESULTS of TELLUROMETRIC EXPERIMENTS IN ISRAEL

TSVI MEIDAV and ALEXANDER KLANG

AUGUST 1965

TM/390/65

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[40501]

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ABSTRACT

Natural earth currents have taken increasingly greater role in geophysical exploration of large sedimentary basins. The method is based on the principle that the natural current sheets tend to concentrate in the sedimentary strata, and hence their relative density and potential drop over these strata reveal the changes in thickness of the sedimentary columns. Recent developments in theory and field technique elsewhere reveal that simultaneous recording of the magnetic field variations above the surface of the earth permit a quantitative determination of thickness of electrically different strata.

The present series of experiments established the feasibility of the telluric technique in Israel, and resulted in mastery of appropriate field and analysis technique which permits the conversion of the tellurograms into geophysical cross sections. Specifications for suitable electrodes and recording equipment are enumerated, and recommendations for the further development of the technique are listed.

GENERAL ASPECTS AND INITIAL RESULTS OF TELLUROMETRIC EXPERIMENTS IN ISRAEL

1.0 STATEMENT OF PURPOSE

In view of the rapid developments in recent years in the understanding of the properties of natural earth currents and the related geomagnetic pulsations, it was decided to try to harness these phenomena for geophysical exploration purposes of great depth extent, by developing appropriate recording technology and suitable interpretation techniques.

The problem of determination of the deeper features of sedimentary basins in Israel, as well as some gross differentiation of the electrical conductivity properties within the sedimentary strata, is of considerable interest to the basic understanding of structural and stratigraphic processes, as well as for applied petroleum prospection.

It was therefore decided to attack the problem in two separate phases :
(a) The present one, in which the basic properties of the telluric currents should be determined at some representative locations in Israel, and rudimentary field and office techniques be developed, and
(b) based upon the findings of the present phase, any modifications of the telluric current recording equipment will be carried out, and the specifications of the required magnetic recording instrument be determined.

In the next phase of operation, a full fledged magneto-telluric prospecting scheme is envisaged, where both electric and magnetic variations will be recorded. The relationship of magnetic to electric pulsation amptitudes as a function of pulsation period, should permit a quantitative determination of depth and resistivity of strata to a depth of many kilometers. The telluric current technique with which we have experimented so far, permits only a qualitative determination of change in depth to the basement, and requires correlation with either routine electrical resistivity soundings or other data for conversion into absolute depths.

The magneto-telluric technique may prove useful also for crustal studies as well, by selecting long pulsation periods which are characteristic of deeply penetrating electric waves.

2.0 THEORY OF TELLUROMETRY

2.1 Historical. After the discovery of electromagnetic induction, Faraday correctly concluded that the variations in magnetic field intensity above the surface of the earth will induce a fluctuating current in the ground. Much later, in 1847, Barlow, a British engineer, made the first attempt for a systematic study of natural earth currents which proved a nuisance in telegraphy transmission over long lines, especially during strong Aurora Borealis appearances. The first attempt for a systematic use of these fluctuating earth ('telluric') currents was made by Conrad Schlumberger in 1912. Schlumberger eventually (1921) concluded that a simultaneous observation of telluric current fluctuations at two separate locations permits a qualitative, or even a semi-quantitative determination, of the subsurface structure. The telluric current method enjoyed considerable eminence among the geophysical techniques employed by French geophysicists in large scale structural investigations, and was used to a lesser extent in other countries as well. In the period of 1941-1948, for example, the French made 21,000 telluric current determinations.

An important development in telluric current techniques came about in the early 1950's, when Tikhonov (1950) in Russia, and Cagniard (1953) in France independently formulated the interrelationship between magnetic induction, and induced electric current in the ground, as a function of frequency, stratification and resistivity of the strata in the ground. It is this latter development which brought about a flurry of theoretical and technological activity in which some of the larger geophysical organisations in many countries are currently engaged. The aim of the current research is to develop a fully quantitative method for determination of the true depths and resistivities of individual strata. Our present study is a first step towards materialization of that goal.

2.2 Origin and nature of the telluric currents.

Natural earth currents are strongly associated with fluctuations of the magnetic field above the surface of the earth. It is generally assumed that the primary cause of the daily magnetic variations lies in the ionosphere. Earth currents display a 27 day recurrence typical of magnetic activity. These currents are of non-uniform

character, and they fluctuate rapidly in direction and time. Gish (1936) plotted the vector direction of these currents at a given instant of time in a dozen observatories scattered in many countries. He found that the resulting vectorial plot describes eight current whorls, fixed in space (Fig. 1). The earth rotates through these whorls, which are more or less symmetrical with respect to the equator.

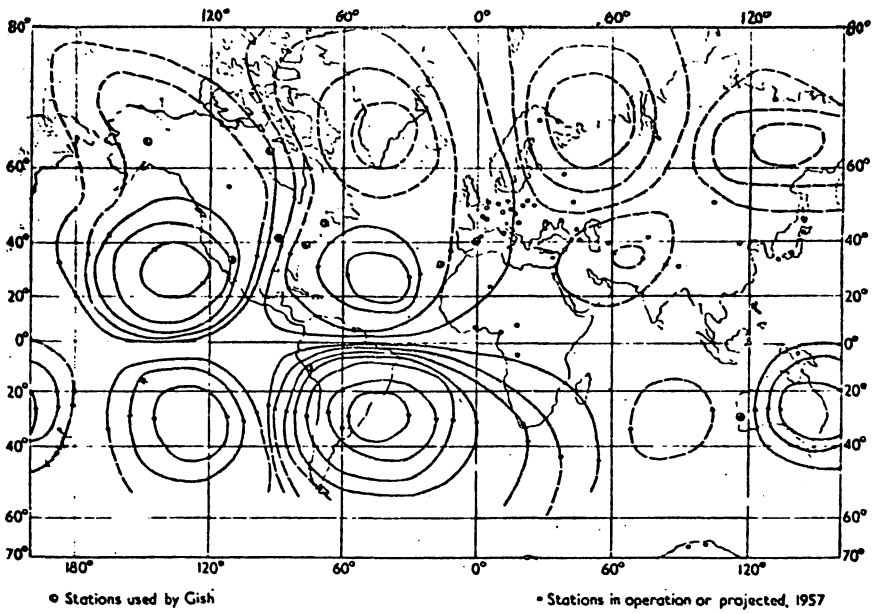


Fig. 1 World system of earth currents (after Gish)

Kunetz (1957) has shown that these currents are remarkably uniform, even in the short period range (one minute or less), over a world wide network of simultaneous recording (Fig. 2).

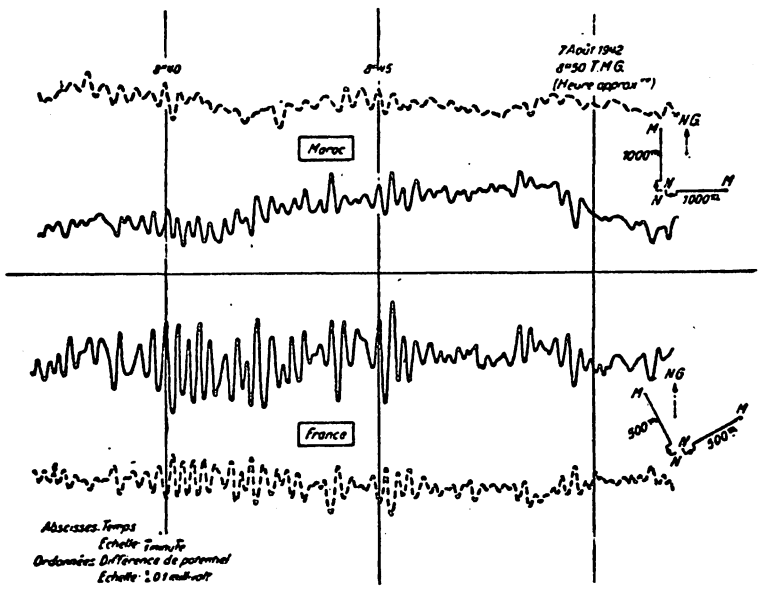


Fig. 2 Simultaneous telluric current variations on Morocco and France.
(After G. Kunetz and M. Schlumberger)

Further studies showed that these remarkably large fluctuating current sheets have persistently correlatable characteristics over entire regions even in the very short period of a few seconds. The French, who first exploited these currents for applied purposes, termed them 'telluric currents', and henceforth they will be referred to by that name.

The general principle of the prospecting method was to record simultaneously current or voltage fluctuations at a base station and a field station. The change in relative amplitude for the same event was used for qualitative or sometimes semi-quantitative determinations of depths to the crystalline basement. This was based on the approximation that most of the telluric current flows in the more conductive sedimentary strata, shunning the much more resistive basement. Hence, an increase in current density at the surface was taken to signify a shallow basement, and conversely a decrease in current density indicated a proportionate increase in depth to the resistive basement.

In our brief experimental study of the present phase, we have achieved, a measure of mastery of the technology of field instrumentation and analysis of telluric current records, as a preparation for the more ambitious program of magneto-telluric prospecting. The following pages contain an exposition of instrumentation and experiments carried out to ascertain that the field records are valid and reliable, or that the random noise could be minimized to below significant levels.

3.0 INSTRUMENTATION

3.1 Recording Equipment

The recording equipment consisted of two dual channel Nanographs, manufactured by SEFRAM, Paris. The SEFRAM recorders are characterized by robust galvanometers, immersed in a liquid of high specific gravity, which made them resistant to considerable shocks. The magnification is of the photoelectric spot-follower type, which yields high current or voltage magnification, with only 20 Watt power consumption. The unshunted galvanometer has a 22 ohm resistance and will deflect 200 mm at 1.5 microamperes or 0.2 millivolts. Shunts of various resistances, permit a selection of lower sensitivities. The inscription was with ink. A variety of paper speeds was made possible by a combination of changeable gears and a speed selector. The paper speed could be varied between 1.2-300 mm/min. A one-minute time marker served as a check on paper speeds. We have introduced into the pen a 50 Hertz vibration to prevent stickiness in writing.

3.2 Control unit.

In addition, a control and calibration unit was constructed and attached to each recorder. The control unit provided for the following functions :

1. Measurement of the resistance of the total external circuit, encompassing wires, contact resistance of the electrodes and the ground.
2. Potentiometers to equalize the external resistances to some arbitrary resistance which was the same at the base and the field station.
3. To provide for a counter e.m.f. to buck out the electrode polarization potentials, which caused a drift of pen to one edge of the paper or the other.

4.0 EXPERIMENTATION WITH ELECTRODES

Although seemingly a mundane matter, the choice of electrodes and their manner of hook-up to the recording equipment proved to be a critical point in attempting to achieve noise-free records. 'Noise' is defined here as any extraneous potentials generated by the sudden disturbance of some electrochemical equilibrium in the ground by planting electrodes in it. As no two electrodes driven in some ground will generate the same response, a troublesome phenomenon of non-similarity had to be overcome.

Experiments with pairs of copper-copper sulfate electrodes driven parallel to each other showed that the equilibrium in the ground, expressed by identity of response of two parallel lines, was achieved only after about six hours, all promises as to the immediacy of stability stated in standard textbooks notwithstanding.

Obviously, such a long time for stabilizing the contact effects, was too long compared with the time needed for the actual recording - half an hour to an hour - and hence shortening of the electrode stabilizing time became vital.

A suggestion by Martel & Meunier (1963) indicated that other electrodes may be less noisy and possess a shorter stabilizing time than the copper-copper sulfate type. The series of experiments enumerated below describes the experiments to achieve noiseless, quickly-stabilized electrodes, and the conclusions of each experiment.

A very strong slow-drift polarization potential existed at the start between all electrodes, causing the entire trace to drift to the end of the paper. This necessitated watchful monitoring by the operator, and the constant bringing back of the trace into scale, by using a bucking current, i.e. current of opposing polarity, fed into the recorder through a trimming potentiometer. The addition of a large electrolytic capacitor (500 μ F) achieved the desired stabilizing effect and tended to keep back the trace perfectly centered (Fig. 3).

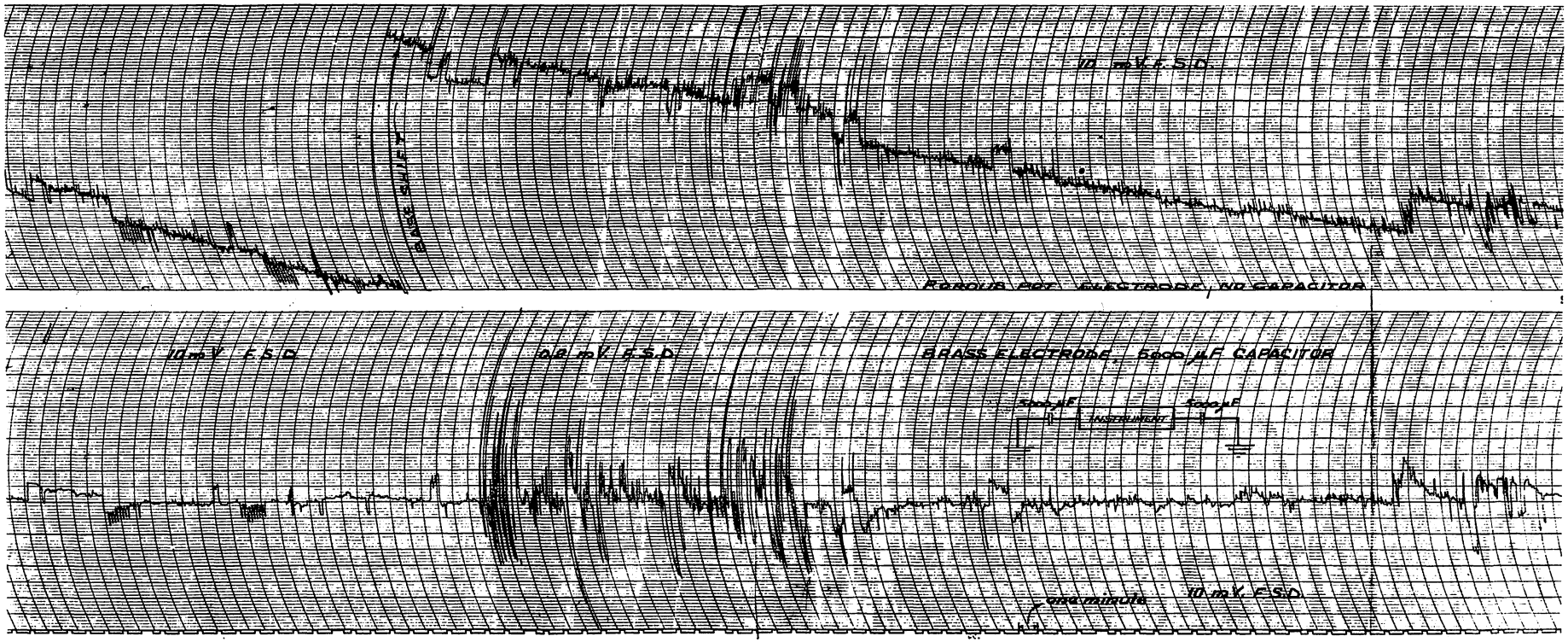


Fig. 3 The drift of the trace without a series capacitor (upper trace) is removed when a capacitor is present (lower trace).

The first test in any series of testing was designed to verify that the recording devices possess an identical response under field conditions. This test was achieved by comparing the response of two channels of the same instrument or of the two instruments when connected to two parallel pairs of electrodes whose contact resistances are equal. For this test, the electrodes were planted in the ground a considerable time (many hours), before the test, so that electrochemical stability was attained, and hence no spurious potentials affect any of the electrodes. To the extent that the contact resistances were not the same, they were equalized by means of the control unit, because the low input resistance of the particular equipment on hand made it quite sensitive to small input impedance differences. The results of such a test are shown in Fig. 4, indicating the high degree of similarity obtainable.

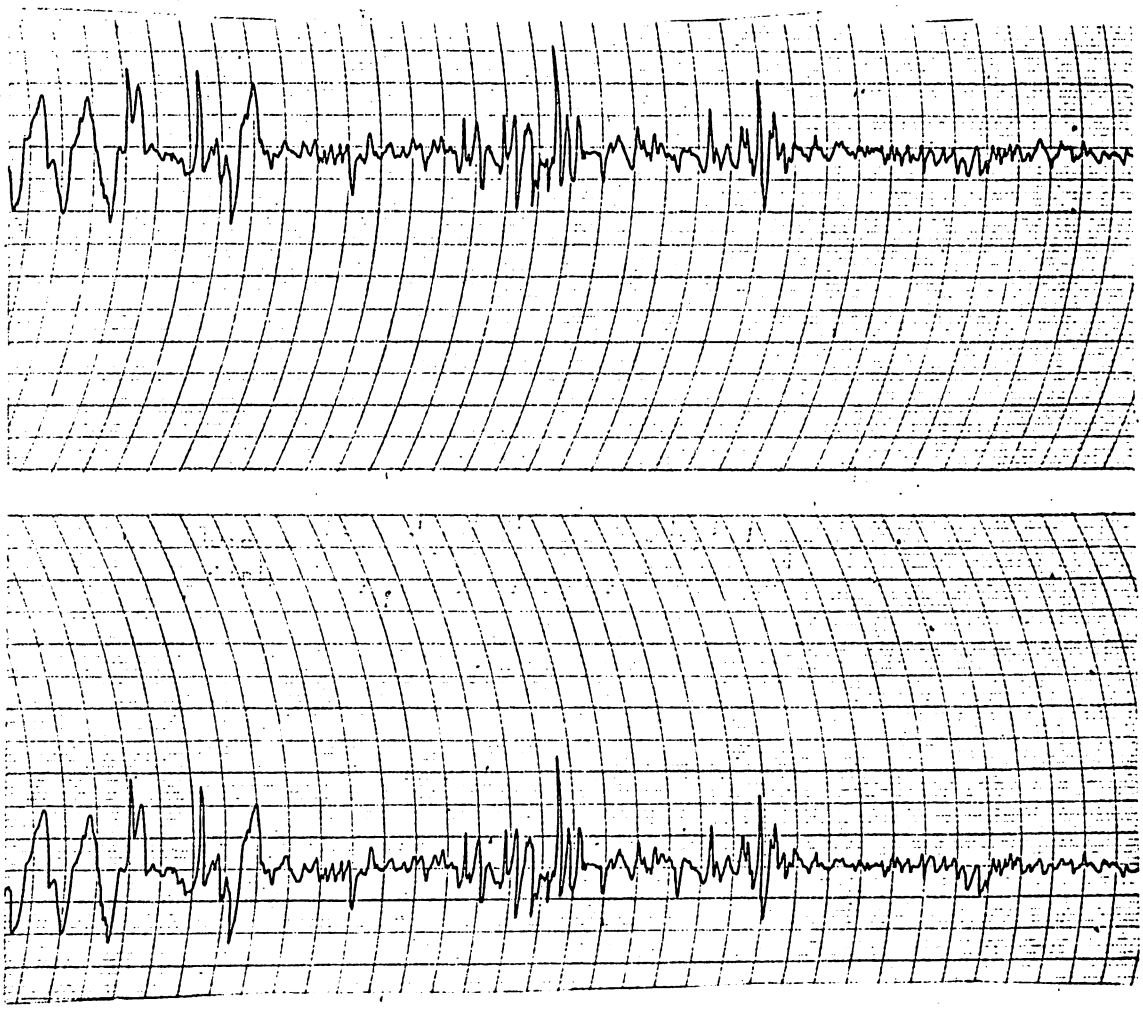


Fig. 4 Similarity test of two channels. Two pairs of electrodes were planted parallel to each other, and the total external resistance of each pair equalized to the other. The response in the two channels is identical.

Experiments have shown that under some unusual circumstances, such as in an industrial zone, or working on the sea beach parallel to the shoreline, strong fluctuating potentials are generated even between electrodes separated only a few centimetres apart (Fig. 5). Such potentials, although small, may make it impossible to achieve a complete identity of response in similarity tests.

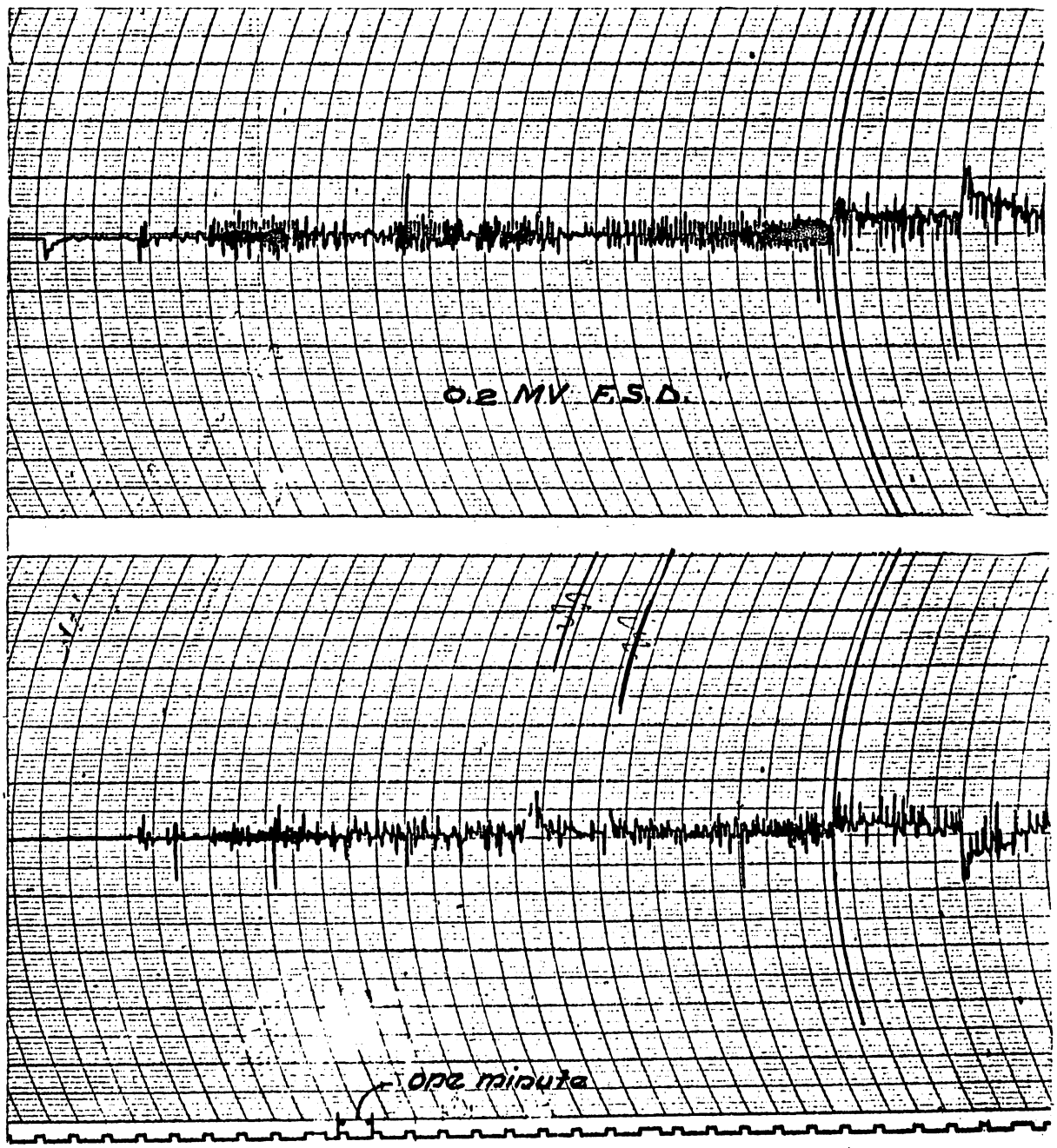


Fig. 5 Noise between electrodes at infinitesimal distance in an industrial zone. Brass electrodes plus capacitor.

Meunier (1962) proposed the use of special porous pot electrodes which according to him generate infinitesimal spurious potentials. The standard copper - copper sulfate electrodes, according to him generate considerable potentials due to the electrofiltration effect through the porous pots. He offered a remedy by immersing the copper sulfate solution in clay, which slowed down the electrofiltration rate, and hence reduced the spurious potentials to a minimum. Our experiments did not verify that assertion completely. Although some improvement was possibly obtained by the use of clay-saturated copper sulfate, the stabilizing time was found to be much longer than that of the brass electrodes. The brass electrodes were usually soaked with salt water and tended to indicate similarity of response within minutes after planting in the ground. Figure 6 shows the records obtained after 5, 30, 70 and 120 minutes after planting for both porous pots and brass electrodes. Figure 6 shows that whereas the brass pairs gave almost identical response within 5 minutes after planting, the porous pots improved only gradually, achieving the same degree of similarity between themselves only 120 minutes after planting. It should be noted that only after 30 minutes do the brass electrode records and the porous pot records start looking alike, and only after an hour do all four records appear as almost identical.

A number of repetitive tests of the brass electrodes, coupled through a capacitor to the recorder, showed that they are the most suitable of all types of electrodes tested. Figure 7 shows a sample of such an experiment, where two sets of arrays were planted about 200 meters apart, each pair orthogonal to the other. The two arrays were parallel. The records show that identity of response in the two arrays has been achieved within minutes after planting in the ground. Such a short duration of time is considered practical. Hence, the series of experiments indicated that the most suitable electrodes were the brass electrodes, preferably drenched in salt water, connected through a large (electrolytic) capacitor in series to the recorder.

The occasional differences in amplitude in some of the similarity tests are attributed to the sensitivity of the present setup to the external resistance, due to the low internal resistance of the instrument. It is possible to overcome that difficulty by constructing a high impedance input device for each channel, which will free the operators from the need of balancing external resistances altogether.

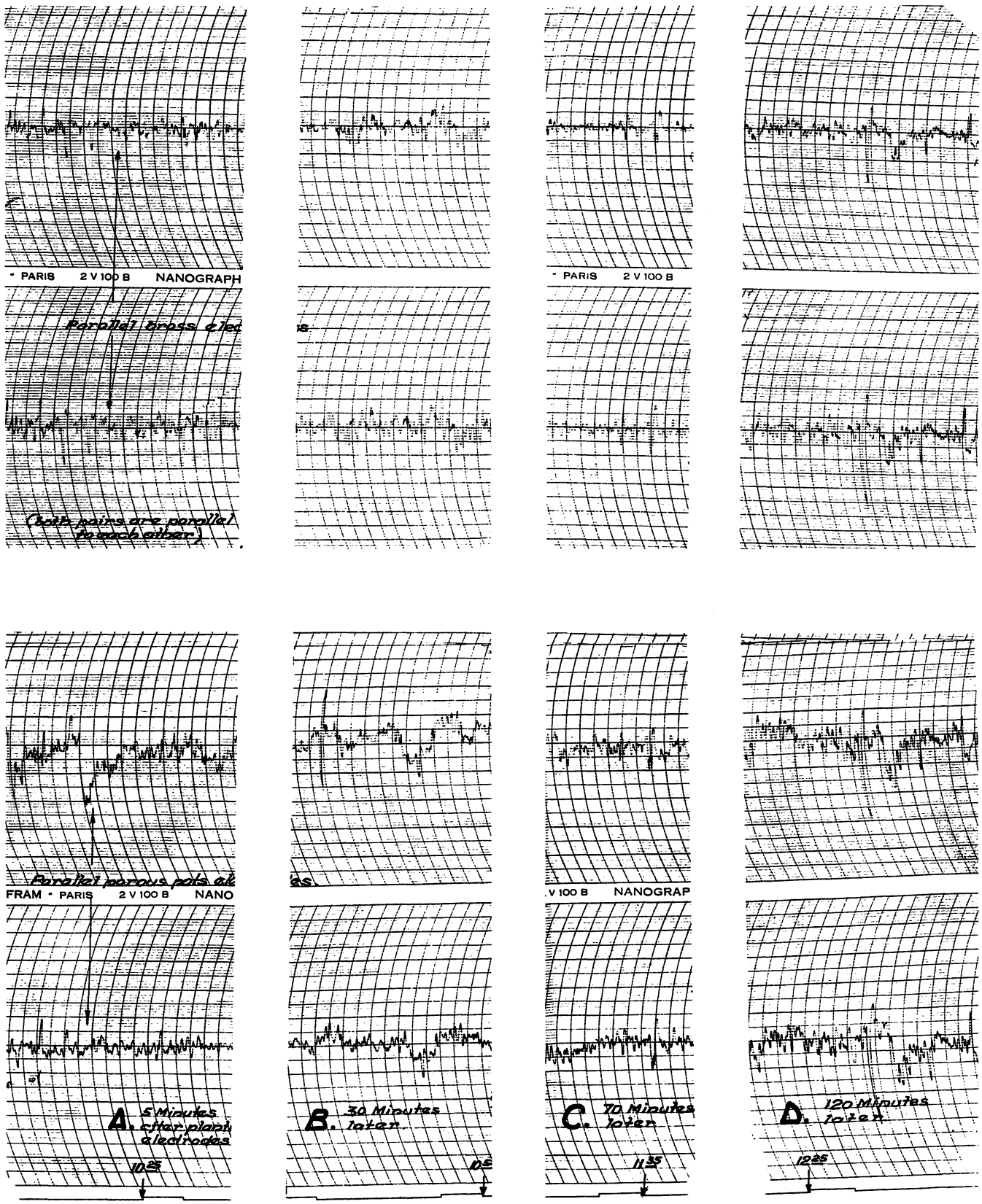


Fig. 6 Comparison of equilibrium time of different electrode types. All electrode pairs are parallel

A) 5 minutes after planting. Porous pot electrodes are not similar among themselves. Brass electrodes are slightly better.

B) 30 minutes later. The brass pair is similar within itself. The propus pots are not.

C) 70 minutes later. All 4 pairs are almost similar.

D) 120 minutes later.

All 4 pairs are more or less identical.

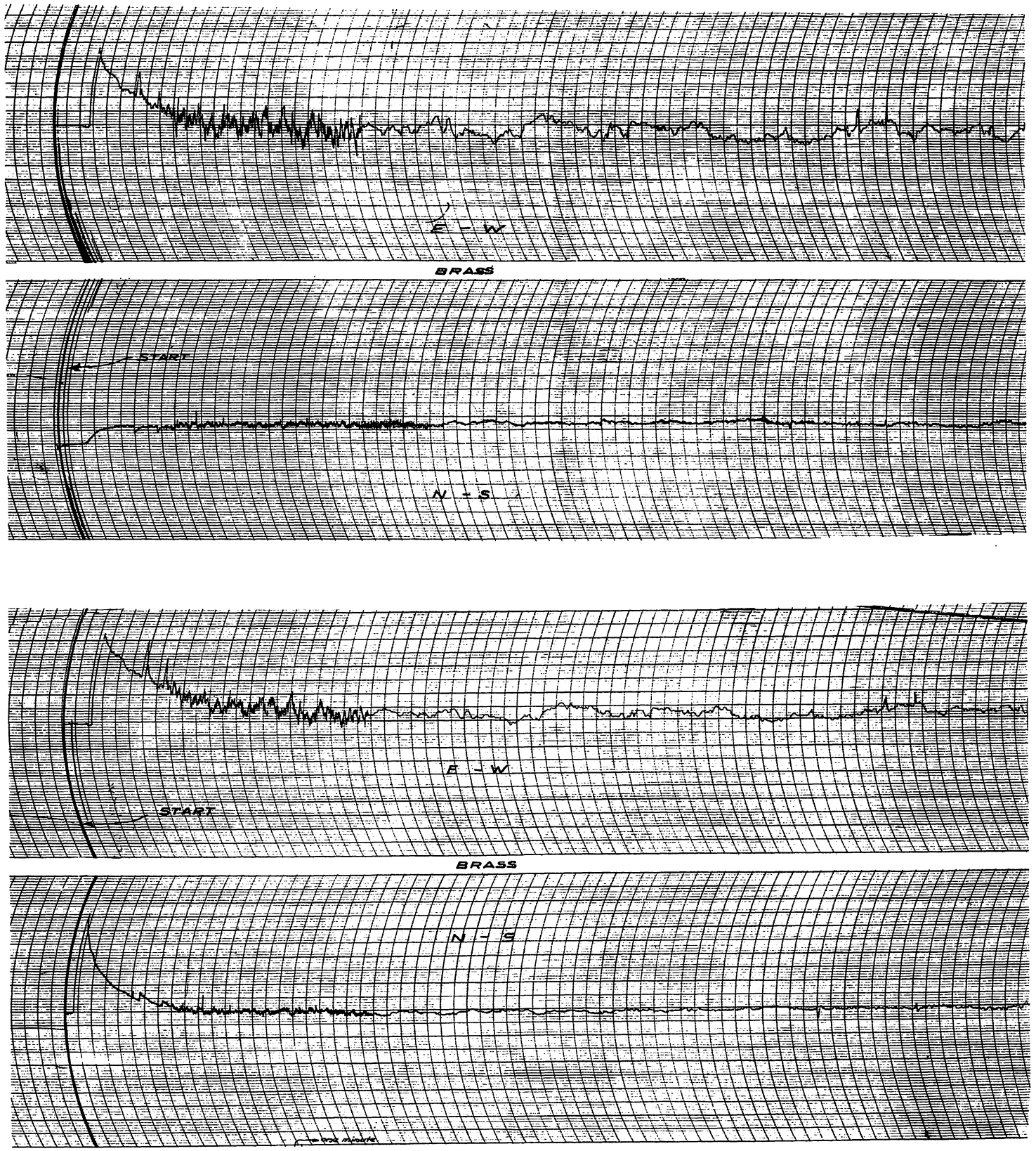


Fig. 7 Comparison test of two sets of parallel arrays. The electrodes are brass, connected through a 5000 μF capacitor to the recorder. Note the similarity almost from the start.

5.0 AMPLITUDES OF THE TELLURIC FIELD IN ISRAEL

It is impossible to draw definite conclusions about amplitude distribution of the telluric field in Israel, for a number of reasons : (1) Most of the experimentation was carried out in a geographically limited area, the northwestern part of the Negev, (2) It is quite conceivable that in different seasons or even in different days, the amptitude of the telluric field could change considerably, (3) The present series of experiments has been carried out in a year of quiet solar activity.

There is no geomagnetic or telluric observatory in Israel as yet. Such an observatory, if and when it is established, will be able to offer important data regarding the variations of electrical and magnetic fields at the surface of the earth.

Typical values of the telluric field in the northwestern Negev, obtained through recording on two consecutive days, are presented in Fig. 8.

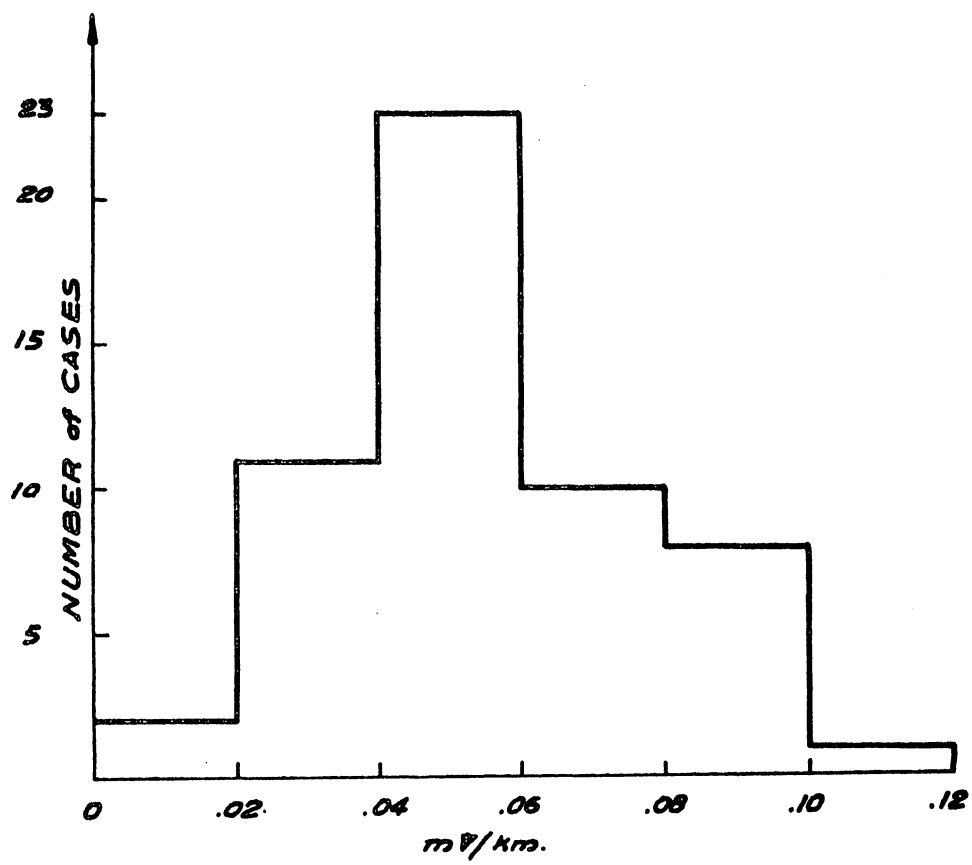


Fig. 8 Histogram of the amplitudes of the telluric field in the northwestern Negev (Ofakim - Pattish - Urim Area).

The histogram shows that the modal values of the telluric field during that period were on the order of .04-.06 mv/km. These are quite low values, and indicate either a great thickness of low resistivity sediments, or low magnetic-telluric activity at that date, or possibly that the telluric field is quite low at the latitude of the northern Negev, or any combination of the above factors.

The low telluric amplitude indicates that recording instruments, both telluric and magnetic, should possess high amplification capacity. The requirements of the amplification are discussed in Section 8.0 ("Recommendations" p. 21)

6.0 RESULTS OF THE EXPERIMENTAL PROFILES

6.1 Negba-Kokhav area

An experimental telluric profile was surveyed in the Negba-Kokhav area where a number of drill holes furnished a fairly complete geological cross-section.

The geological information shows that, in that area, the Cretaceous formations are covered by the Saquia formation, the thickness of which is 200 - 400 m. The existence of an anticlinal structure is verified by the Heletz wells, which are situated a few kilometers to the south .

The electrical soundings in the area revealed low values of the resistivity, both for the Saqia and for the Cretaceous formations. The apparent resistivities to a depth of 1.5-2 kilometers are of the order of 10-20 ohm-meters.

Four stations were occupied along that profile (Stations 1 and 3 with base B and Station 4 with base 3) (Fig. 9).

The tellurograms which were obtained are greatly affected by interferences and the general amplitude of the telluric field is very low. The interferences may be partly due to the drilling activity in the area. The low values of the resistivity are a possible explanation of the low amplitudes of the telluric field; the great depth to the basement complex may be another reason for the low amplitudes.

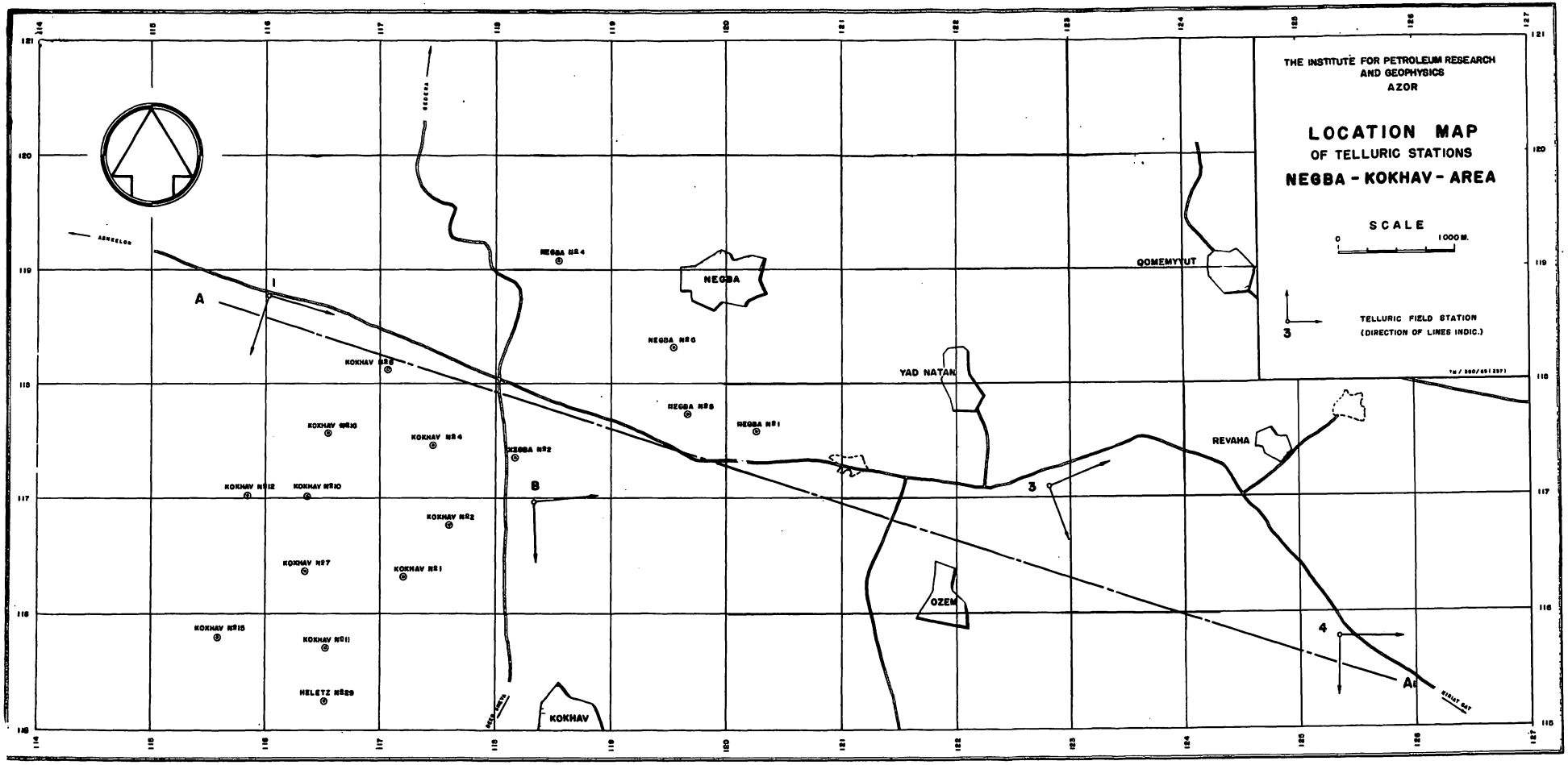


Fig. 9 Location map of telluric stations. Negba-Kokhav area.

The profile was interpreted by calculating the parameter K and the ratios of the telluric field intensities. Assuming an arbitrary value of the depth of the basement at the base station, a cross-section of basement profile may be constructed, as shown in Fig. 10.

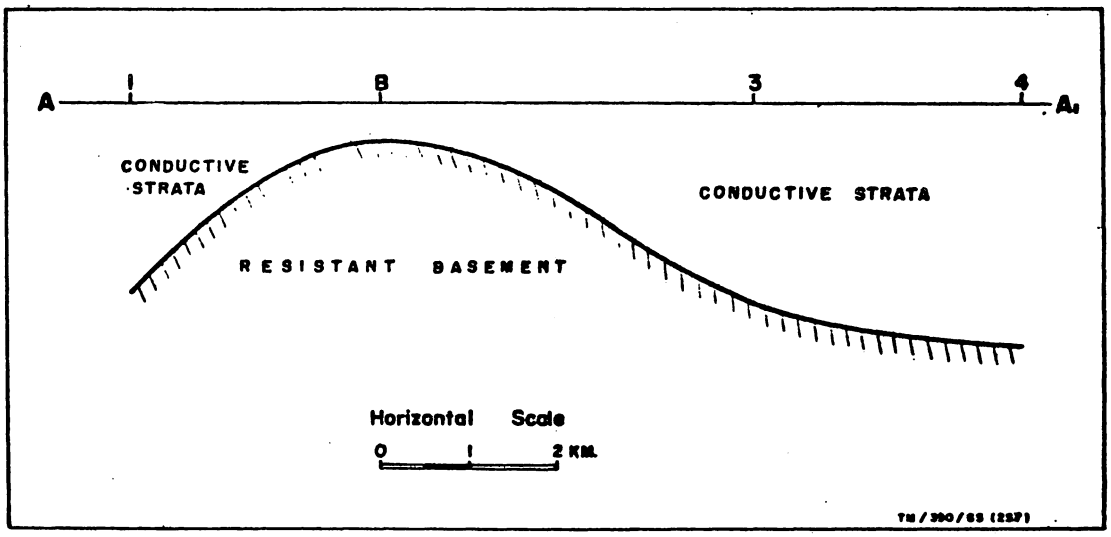


Fig. 10 Interpreted telluric profile, Negba-Kokhav area. The resistant basement structure corresponds to the anticlinal structure of the area.

The layer termed here 'basement' cannot be defined as yet, geologically or depthwise. The least that we know is that the electrical basement is deeper than Cretaceous strata, because Cretaceous rocks in that region possess low resistivities. When either magneto-telluric soundings or very deep electrical resistivity soundings are carried out in that region, it will be possible to identify the geological significance of that electrically resistant basement.

It is interesting to note that the electrical basement high corresponds to the axis of the Heletz anticlinal structure.

6.2 Ofakim - Pattish - Urim area.

A second experimental profile with the telluric method was surveyed in the Ofakim-Pattish-Urim area. This area was chosen to try out the method in a noise free area. There are no wells in the area, the nearest, Nirim No. 1, being about 4 kilometers to the east. (Fig. 11).

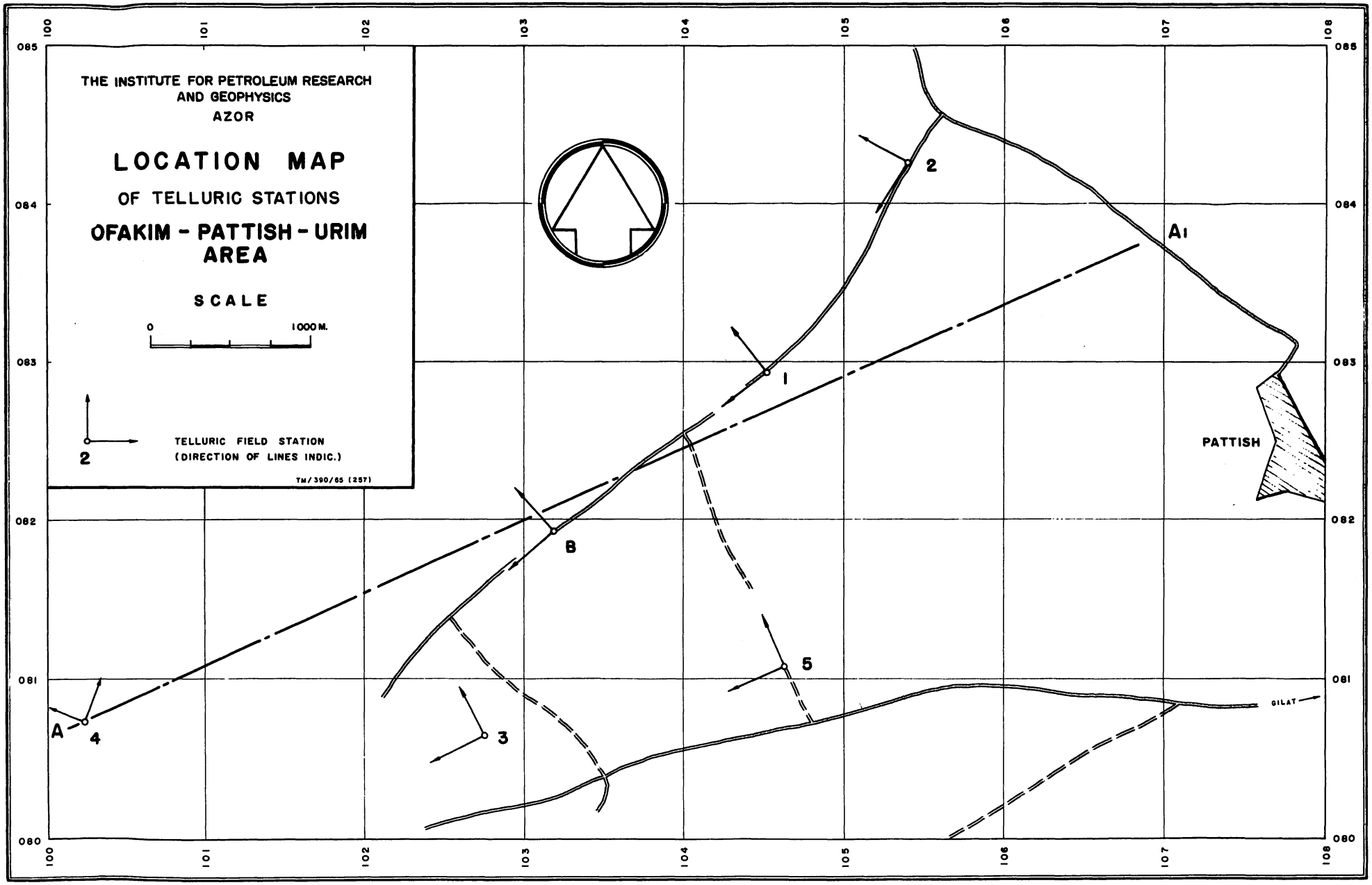


Fig. 11 Location map of telluric stations, Ofakim-Pattish-Urim area

The general geological information indicates the presence of the Cretaceous formations at low depths (300-500 m.). In this area, as in Negba-Kokhav, the apparent resistivity to a depth of two kilometers is probably less than 20 ohm-m.

Measurements were carried out in five field stations simultaneously with a permanent base station. The first two stations could not be used in the interpretation due to low amplitudes of variation of the telluric field. The interpretation of the other three stations was carried out using ellipse and the amplitude method (paramenter μ).

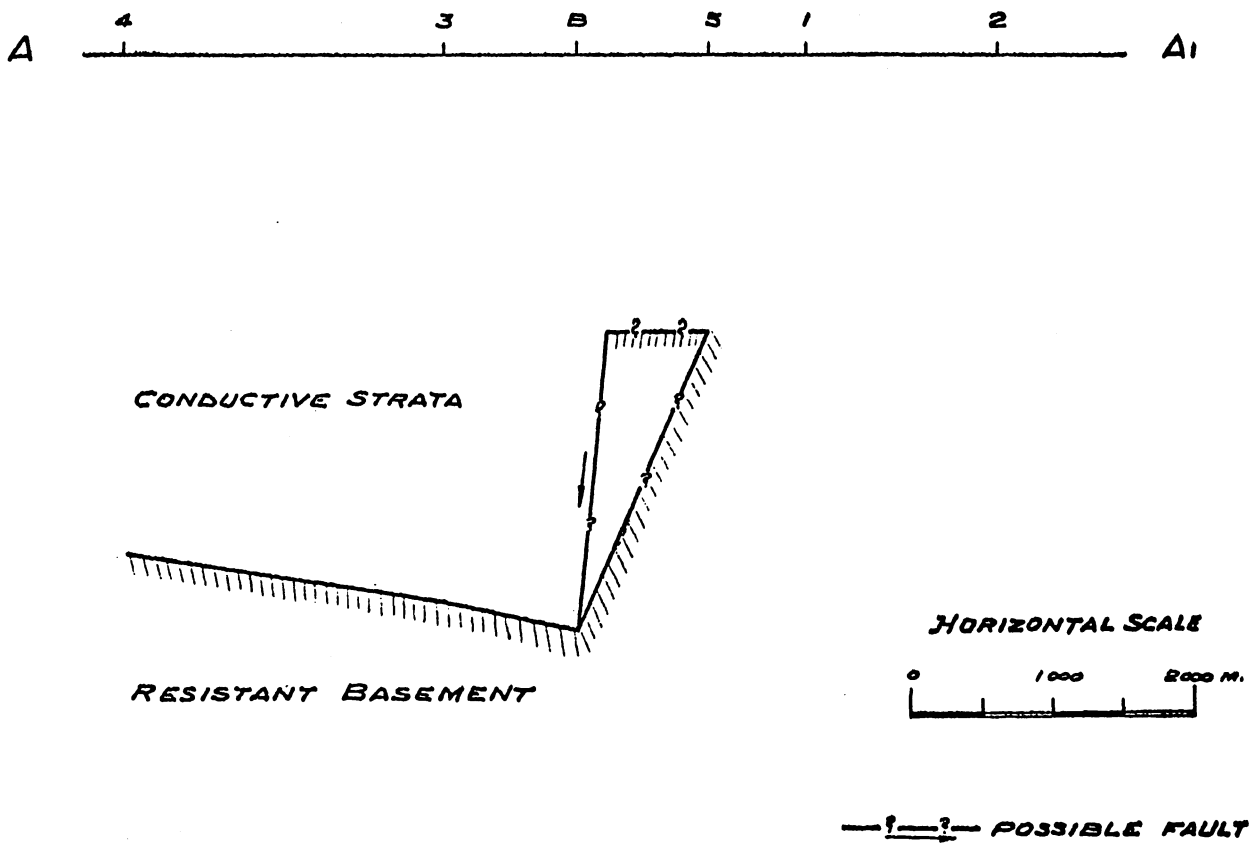


Fig. 12 Interpreted telluric profile, Ofakim-Pattish - Urim area .

Figure 12 is a cross-section of possible interpretations of the telluric profile. No vertical scale can be given at this stage. The relative position of the electrical basement at Station 5 suggests the possible existence of a fault. This interpretation is uncertain as the tellurograms of stations 1 and 2 could not be interpreted, and due to the lack of sufficient geological information. With either instrumental improvement (pre-amplification) or a denser net of telluric stations, a more reliable interpretation could be effected.

Taking in consideration the probable low values of the electric resistivity for the Cretaceous formations and without any information regarding the resistivity of older formations, the position of the telluric (electrically resistant) basement in the geological scale cannot be defined as yet.

7.0 CONCLUSIONS

The results of the present research program indicate that the telluric method, as well as the magneto-telluric technique, which is a derived method, are applicable in Israel.

The telluric current fluctuations during the experimentation were quite weak, necessitating special means for reliable interpretation. Conversely, it is expected that the associated magnetic fluctuations will be quite low, of the order of a few tens of milligammas in the range of periods of interest.

The techniques of obtaining reliable and reproducible field data has been mastered, and analytical techniques of interpretation have successfully been applied to the results of the few experimental profiles.

By adding magnetic variation measurements to the earth current measurements, it will be possible to make full-fledged magneto-telluric measurements. With the added instrumentation and improvements in the present telluric instrumentation, quantitative information regarding depth of different resistivity strata can be obtained to a depth of many kilometers. The magneto-telluric technique should prove of considerable value in large scale geological surveys of sedimentary basins, and should prove beneficial in petroleum exploration schemes.

8.0 RECOMMENDATIONS

1. In order to increase the sensitivity of the present recording systems, it is required to increase the input impedance of the recorders or add pre-amplifiers.

2. For both efficient telluric current recording, as well as facility of analysing magneto-telluric records, an efficient filter system should be added to each channel.

3. A radio communication system between the field station and base station is a requisite for efficient operation, as well as synchronization of time bases and fiducial marks.

4. Two magnetometers of high sensitivity should be added to the present instrumentation to make the application of magneto-telluric methods possible.

9.0 ACKNOWLEDGEMENTS

We wish to acknowledge with gratitude the advice and information furnished to us by Mr. Clerc, Centre d'Etudes Geophysiques, France, and Messrs. Martel and Meurier of France.

The control units were constructed by Mr. Herzl Smocha, our electonic technician, who also carried out the field and laboratory experiments. Mrs. Zafra Vered assisted in carrying out the calculations and Mr. Heinal drafted most of the illustrations. We thank them all for their conscientious work.

A P P E N D I X

ANALYSIS AND INTERPRETATION TECHNIQUES

1.1 METHOD OF COMPUTATIONS

The original data obtained in a telluric current survey consists of tellurograms or visual recordings of the variation of the telluric current field at a base station and field station. The interpretation is based upon the assumption that at any particular moment the components of the telluric current field at the base station are related to the components of the field station by a system of linear homogeneous equations.

If :

E_{Su} = the component of the telluric current field at the field station in the U direction

E_{Sv} = the component of the telluric current field at the field station in the V direction

E_{Bx} = the component of the telluric current field at the base station in the X direction

E_{By} = the component of the telluric current field at the base station in the Y direction

one can write : (Fig. 13) :

$$\left.\begin{aligned} E_{Su} &= a\,E_{Bx} + bE_{By} \\ E_{Sv} &= c\,B_{Bx} + dE_{By} \end{aligned}\right) \qquad (1)$$

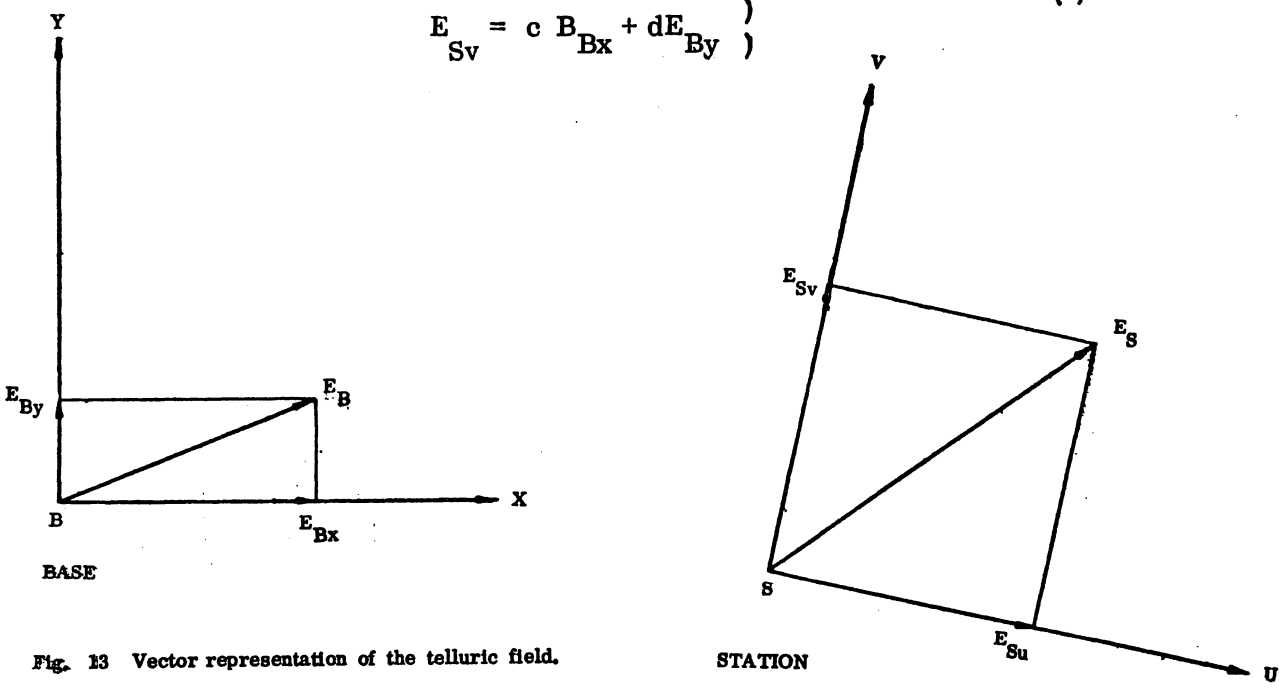


Fig. 13 Vector representation of the telluric field.

It can be shown that if a linear relationship exists between the components of the telluric current field, a linear relationship exists also between the resultant current vectors.

From this :

$$\frac{E_S}{E_B} = \frac{E'_S}{E'_B} = \frac{E_S - E'_S}{E_B - E'_B} \qquad (2)$$

Where : E_S = the resultant vector at the field station at the time T

E'_S = the resultant vector at the field station at the time $T' = T + \triangle T$

E_B = the resultant vector at the base station at the time T

E'_B = the resultant at the time $T' = T + \triangle T$

The expression (2) shows that one can employ the differences of the telluric current field in the same way as the absolute value of the field.

By this, another difficulty can be overcome. In measuring the voltage between a pair of electrodes it is not possible to separate the electrochemical potentials due to the polarisation of the electrodes from the ohmic potential caused by the telluric current field. The trace drawn on the oscillograph is the sum of the two potentials. By measuring the differences of the telluric vector over a relatively short period, so that electrochemical potentials do not change significantly, one obtains differences of the telluric current field.

The choice of the method of interpretation is determined by the manner in which the telluric current is polarised. If the telluric current field is nonlinearly polarised, tellurograms are interpreted using the ellipse method.

In zones where the telluric current field is completely linearly polarised the ellipse is not usable and one determines the amplitude ratios, the parameter μ .

1.1.1 The ellipse method

The interpretation by the ellipse method is based on the following considerations. For a given station, the vector of the telluric current field E_S changes in direction and in amplitude with time. The same is valid for the vector E_B , the field at base, homologous of E_S . Dividing E_B and E_S by E_B , one obtains again two homologous vectors.

$$\frac{E_B}{E_B} = 1 \text{ and } \frac{E_S}{E_B}$$

with the same directions as E_B and E_S

The vector $\frac{E_B}{E_B} = 1$ describes a circle with the radius 1

The vector $\frac{E_S}{E_B}$ describes an ellipse.

Let us define a parameter K as : $K = \sqrt{\frac{\Phi S}{\Phi B}}$ (3)

where ΦS = the area of the ellipse described by the vector $\frac{E_S}{E_B}$ at the station,

ΦB = the area of the circle described by the vector $\frac{E_B}{E_B}$ at the base,

it can be found : $$K = \sqrt{(ad - bc)\frac{\sin \alpha}{\sin \beta}} \quad (4)$$

where a, b, c, and d are the parameters from the equations (1),

is the angle between the measuring lines at the base station,

is the angle between the measuring lines at the field station.

In our case with $\alpha = \beta = 90^o$ we have :

$$K = \sqrt{ad - bc} \quad (5)$$

The parameters a, b, c, d are determined by the distribution of the electrical properties in the medium and they are constants for a station S relative to the base B, for the chosen system of coordinates.

We can find the constants a, b, c, d in two ways :

A. By calculating the values a, b, c, d from equation (1) for a number of pairs of vectors read at closely spaced intervals, after the synchronization of the tellurograms :

$$E_{Su1} = aE_{Bx1} + bE_{By1} \qquad E_{Sv1} = cE_{Bx1} + dE_{By1}$$

$$E_{Su2} = aE_{Bx2} + bE_{By2} \qquad E_{Sv2} = cE_{Bx2} + dE_{By2} \qquad (6)$$

$$E_{Sui} = aE_{Bxi} + bE_{Byi} \qquad E_{Svi} = cE_{Bxi} + dE_{Byi}$$

From equation 6 one obtains by the method of least squares the values of a, b, c, d and one finds the value of K.

B. Another way is to construct the ellipse. This is done as follows :

1) synchronization of the tellurograms,
2) construction of vector diagrams,
3) reduction of the vector diagram and the construction of ellipses.
4) determination of the parameters K and M

$$M = \left[\text{Semi minor axis of the ellipse / Semi major axis of the ellipse}\right]^{1/2}$$

5) determination of the coefficients of the equations relating the telluric vectors at the field and base station, a, b, c, d.

1) Synchronization of the tellurograms

The base station and field station tellurograms are synchronized by aligning simultaneous time marks recorded on both records.

2) Construction of Vector diagrams

Values of ΔE_{Bx} and ΔE_{By} from a base station tellurogram and of ΔE_{Su} and ΔE_{Sv} from a field station tellurogram are read at closely spaced intervals over periods in the range from seconds to hundreds of seconds (see fig. 14). These time intervals must be synchronized at the

base and field station with appropriate time marks. The amplitudes are read in the way shown on fig. 15, considering positive an upward deflection.

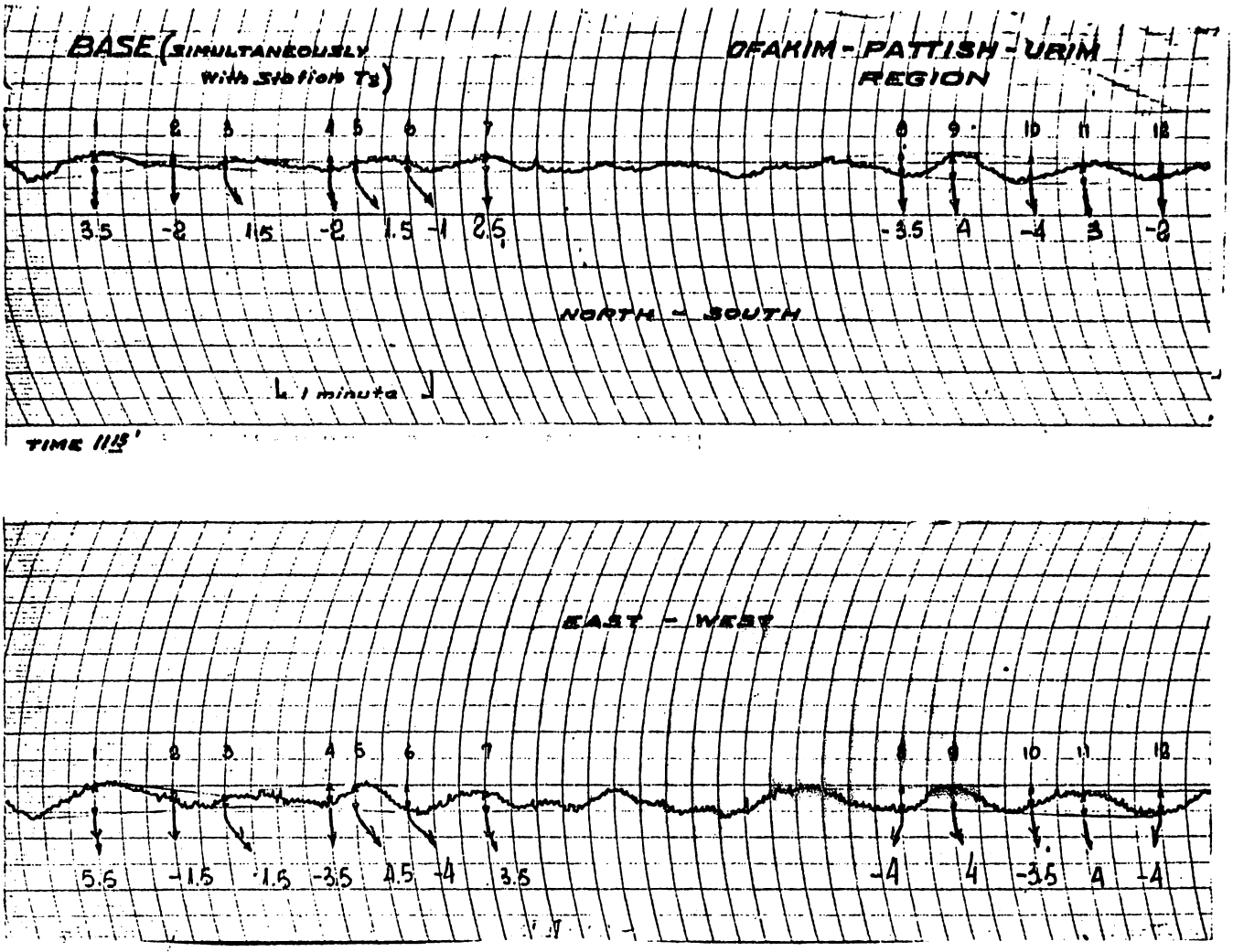


Fig. 14 Field tellurogram. The amplitudes of the telluric vector are read at close intervals for both components.

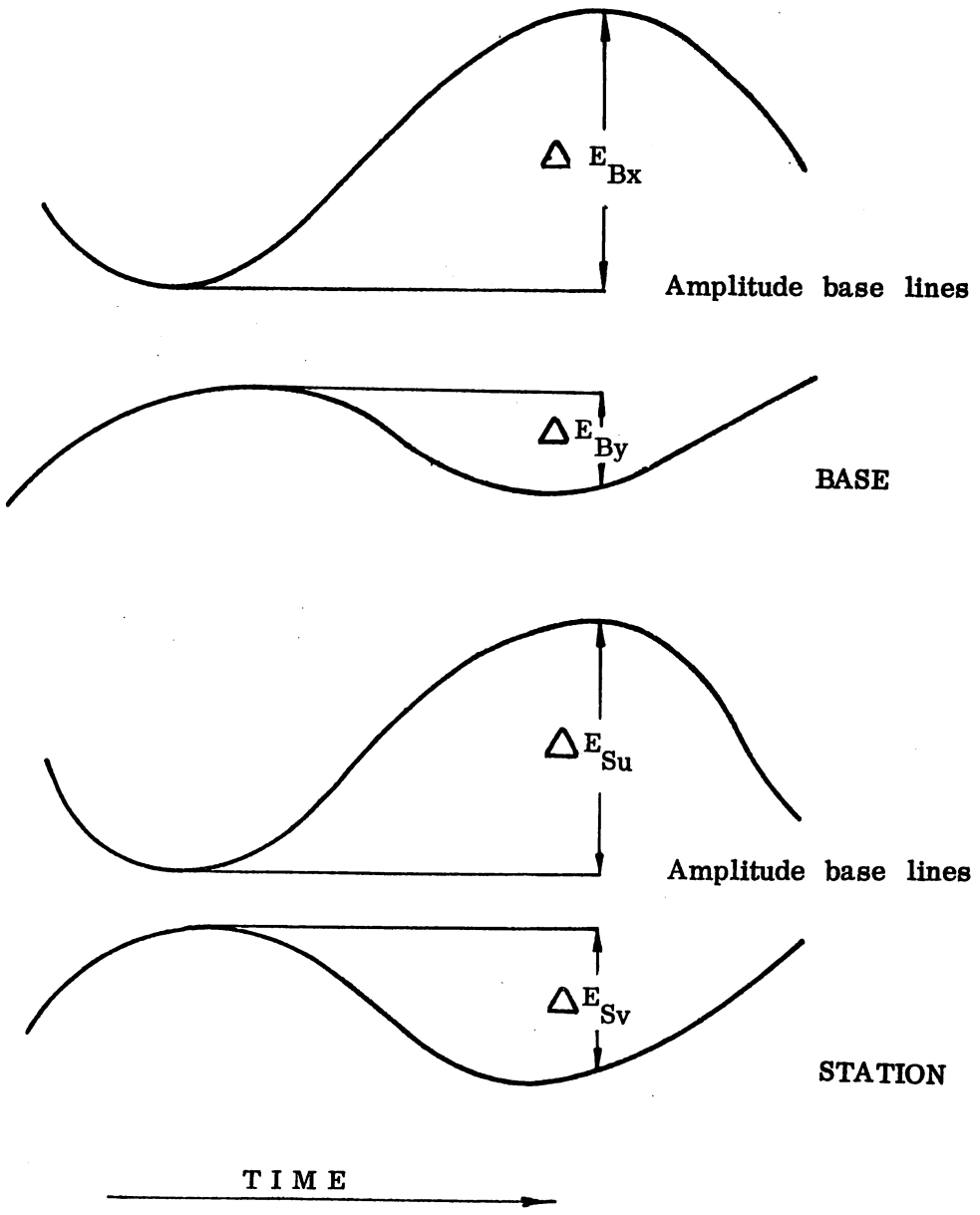


Fig. 15 Method of simultaneous reading the amplitudes of the telluric components.

These are the components of the telluric field vectors at the base and at the station and one can construct the resultant vectors (fig. 13). In this way, drawing the coordinate axes X, Y for the base, and U, V for the station with azimuths corresponding to the true azimuths of the measuring lines of the field, one constructs the telluric field vectors ΔE_B and ΔE_S for base and station.

3) Reduction of the vector diagram and construction of ellipses

The objective in plotting the diagrams of ΔE_B and ΔE_S is to determine an ellipse which will encircle these vectors once they have been normalized. Consider a simultaneous variation in an electric field at a base and field station given by ΔE_B and ΔE_S divided by the same scalar factor. By this, the relationship between the components of the vectors from equation (1) are not changed.

The scalar factor f is selected so that the vector $f \cdot \Delta E_B$ will lie on a circle of radius R.

The conditions are satisfied if

$$f = \frac{R}{\Delta E_B} \qquad (7)$$

The vector at the field station, which corresponds to this one is $f \cdot \Delta E_S$, which lies on an ellipse (according to the theory of linear transformations). This ellipse is the transform of the circle with radius R.

By following the same procedure with a series of vectors it is possible to obtain a series of points lying on the perimeter of the field ellipse. By plotting the reduced vector on the direction opposite the direction of the original vector one obtains reflected points, which lie on the same ellipse. It is a simple matter to construct an ellipse through these points (fig. 16).

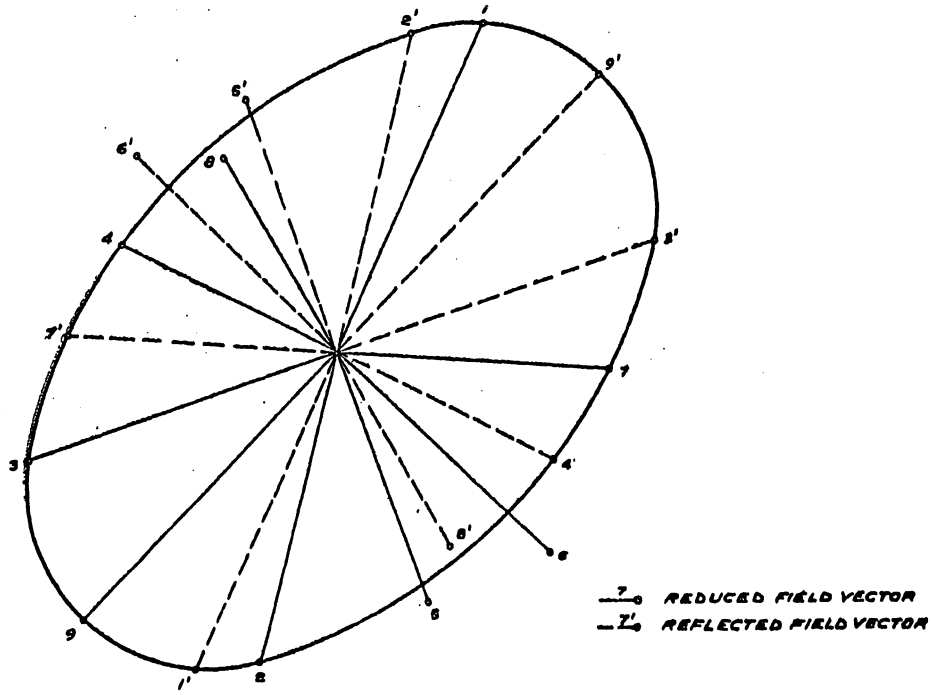


Fig. 16 Construction of the telluric ellipse.

A base station ellipse may be constructed in a similar manner. The scalar h satisfies the equation :

$$h = \frac{R}{\Delta E_S} \qquad (8)$$

With this transformation a field vector h ΔE_S will be obtained, which traces out a circle of radius R, and a base vector h ΔE_B, which traces out an ellipse.

The values of f and h are normalizing parameters and this operation is called reduction of the vector diagram. In this way one can construct also the ellipse of the base station. Having plotted the base and the field ellipse, it is a simple matter to determine the major and minor semi-axes A_B, B_B of the base ellipse and the major and minor semi-axes A_S, B_S of the field ellipse. The semi-axes must satisfy the condition

$$A_B \cdot B_S = A_S \cdot B_B = R^2 \qquad (9)$$

4. Determination of the parameters K and M

The value of K is defined by ;

$$K = \sqrt{\frac{\Phi S}{\Phi B}} \qquad (3)$$

where ΦS = the area of the field ellipse (with semi-axes A_S, B_S)

ΦB = the area of the circle R, the transform of the field ellipse.

Thus we have :

$$K = \frac{\sqrt{A_S B_S}}{R} \qquad (10)$$

Therefore, the parameter K may be determined by measuring the semi-axes of the field ellipse.

The parameter M is determined from the equation :

$$M = \sqrt{\frac{B_S}{A_S}} \qquad (11)$$

5. Determination of the linear coefficients (a, b, c, d)

The linear homogeneous equations :

$$E_{Su} = aE_{Bx} + bE_{By}$$
$$E_{Sv} = cE_{Bx} + dE_{By} \qquad (1)$$

may be solved graphically with the ellipses.

Consider a base ellipse and a field ellipse obtained after the reducing vector diagram operation.

We construct auxiliary coordinates X', Y' and U', V' such that the X'-axis lies along the minor axis of the base ellipse and U' axis lies along the major axis of the field ellipse.

The angle φ, between the minor axis X' and axis X at the base station is measured. We calculated the angles :

$$\varphi_m = \arctan (M \tan \varphi) \tag{12}$$

$$\varphi_n = \left[\arctan M \tan \left(\varphi + \frac{\pi}{2} \right)\right] = -\arctan (M \cot \varphi)$$

The central radii m and n at the field ellipse may now be constructed since they are included at an angle to the major axis of φ_m for the vector m and φ_n, for the vector n. (Fig. 17).

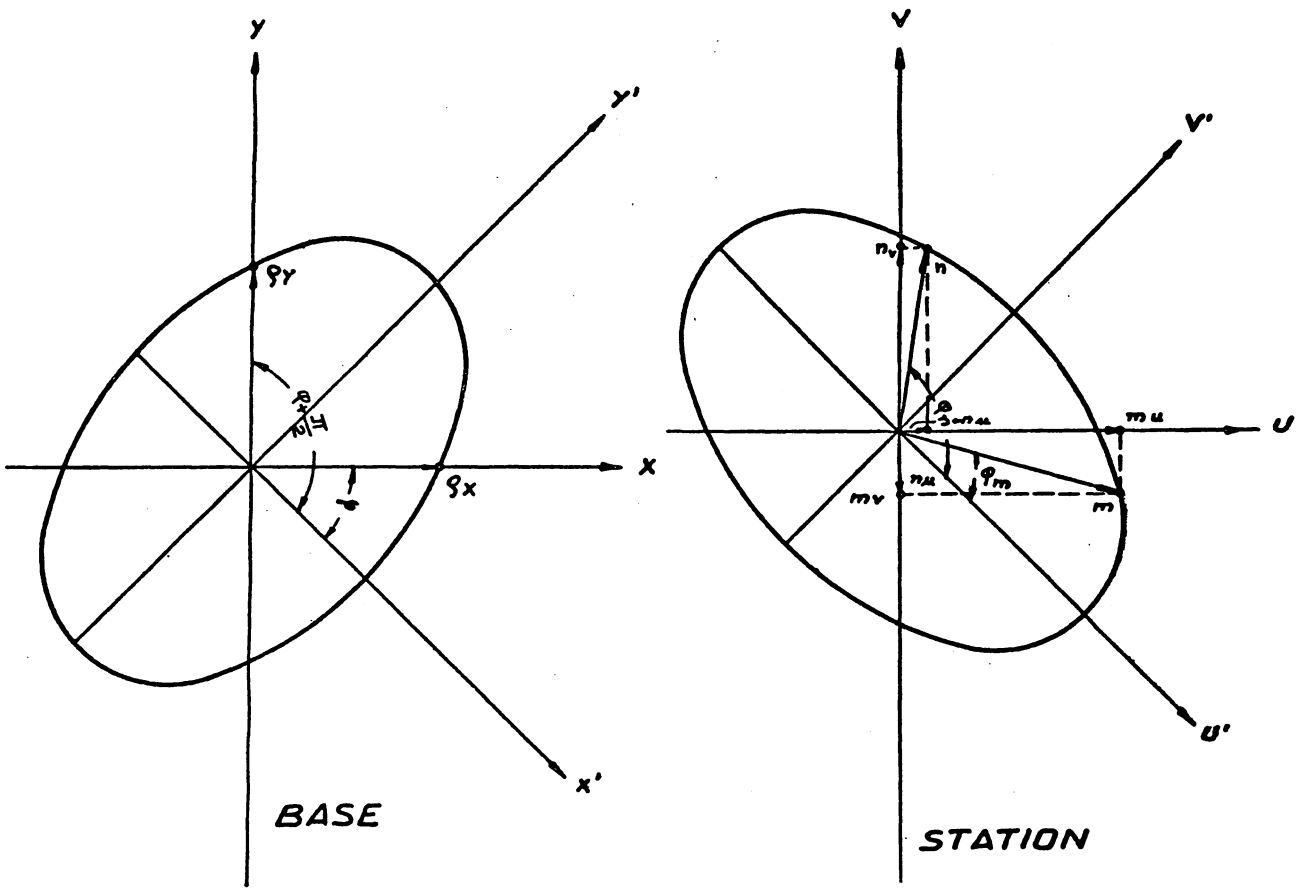


Fig. 17. Construction of vectors m and n ; determination of constants a, b, c, and d.

In constructing the vectors m and n one must observe the following points :

a) The angle between vectors m and n, as measured from the vector m counter clockwise must be less than 180° ;

b) If the minor or major semi-axis of the base station ellipse falls between the positive X and Y axes, the vectors m and n must also be included between the major or the minor semi-axis of the field ellipse.

The length of the vectors m and n should satisfy the equations :

$$1 - \frac{R^2}{m\rho_x} = 0$$

$$1 - \frac{R^2}{n\rho_y} = 0 \qquad (13)$$

where : m, n are the length of the vectors m and n

ρ_x, ρ_y are the central radii of the base ellipse having axial directions X, Y.

To prove that the graphical construction is valid, the following value should be calculated :

$$\delta m = q - \frac{R^2}{m\rho_x}$$

$$\delta n = 1 - \frac{R^2}{n\rho_y} \qquad (14)$$

According to equations (13) both δ m and δn should be 0. In practical cases the slight errors in the measurements will cause values which differ from 0. It should be considered that the vector components m and n are correctly constructed if the absolute value of δ m and δn are not greater than 0.1.

The coefficients a, b, c, d are determined from the projections of the vectors m and n on the coordinate axis in the system U, V, from the equations :

$$a = \frac{m_U}{R}, \quad b = \frac{n_U}{R}, \quad c = \frac{m_V}{R}, \quad d = \frac{n_V}{R} \qquad (15)$$

Where : m_U and m_V are the projections of the vector m on the axis U, V

n_U and n_V are the projections of the vector on the axis U, V.

The inaccuracy of the values a, b, c, and d obtained by this construction is given by the equation :

$$v = \frac{K}{\sqrt{ad - bc}} \qquad (16)$$

Which according to equation (5) must equal unity. In practical cases, if the error is not more than 5%, the results may be considered satisfactory.

We give an example from field measurements which have been calculated by this method. (Fig. 18-21).

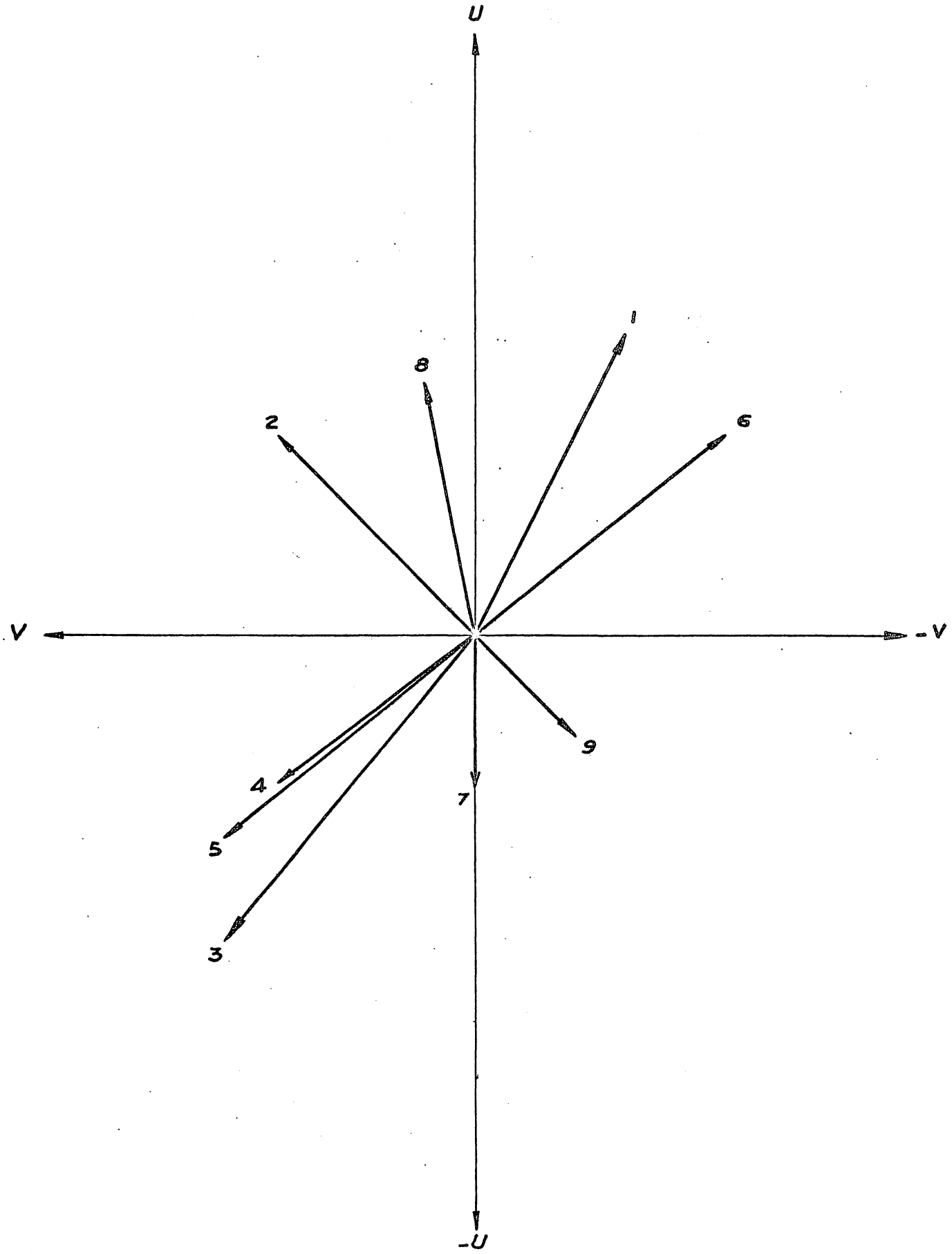


Fig. 18. The raw telluric vector diagram station 4, Urim area.

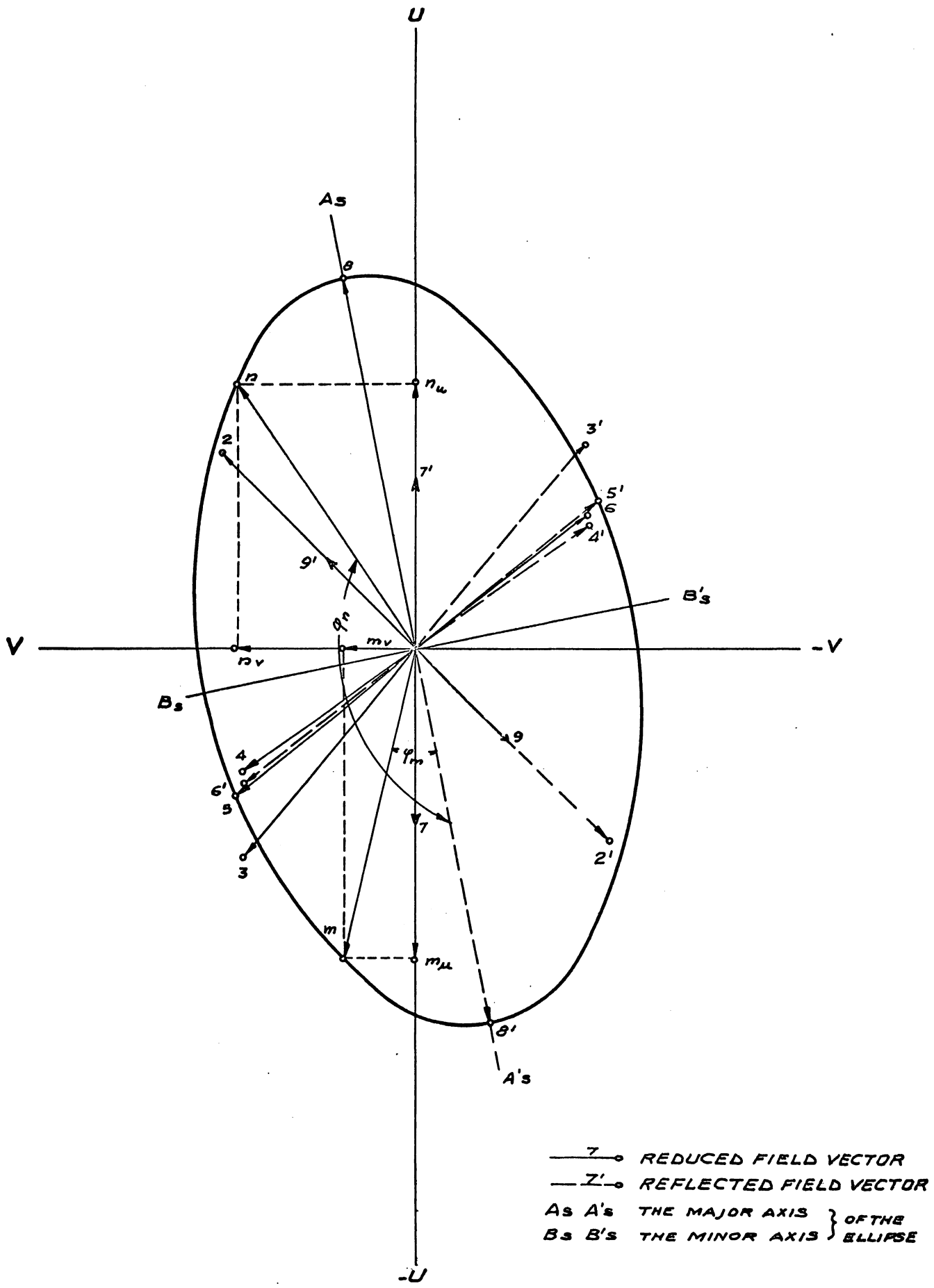
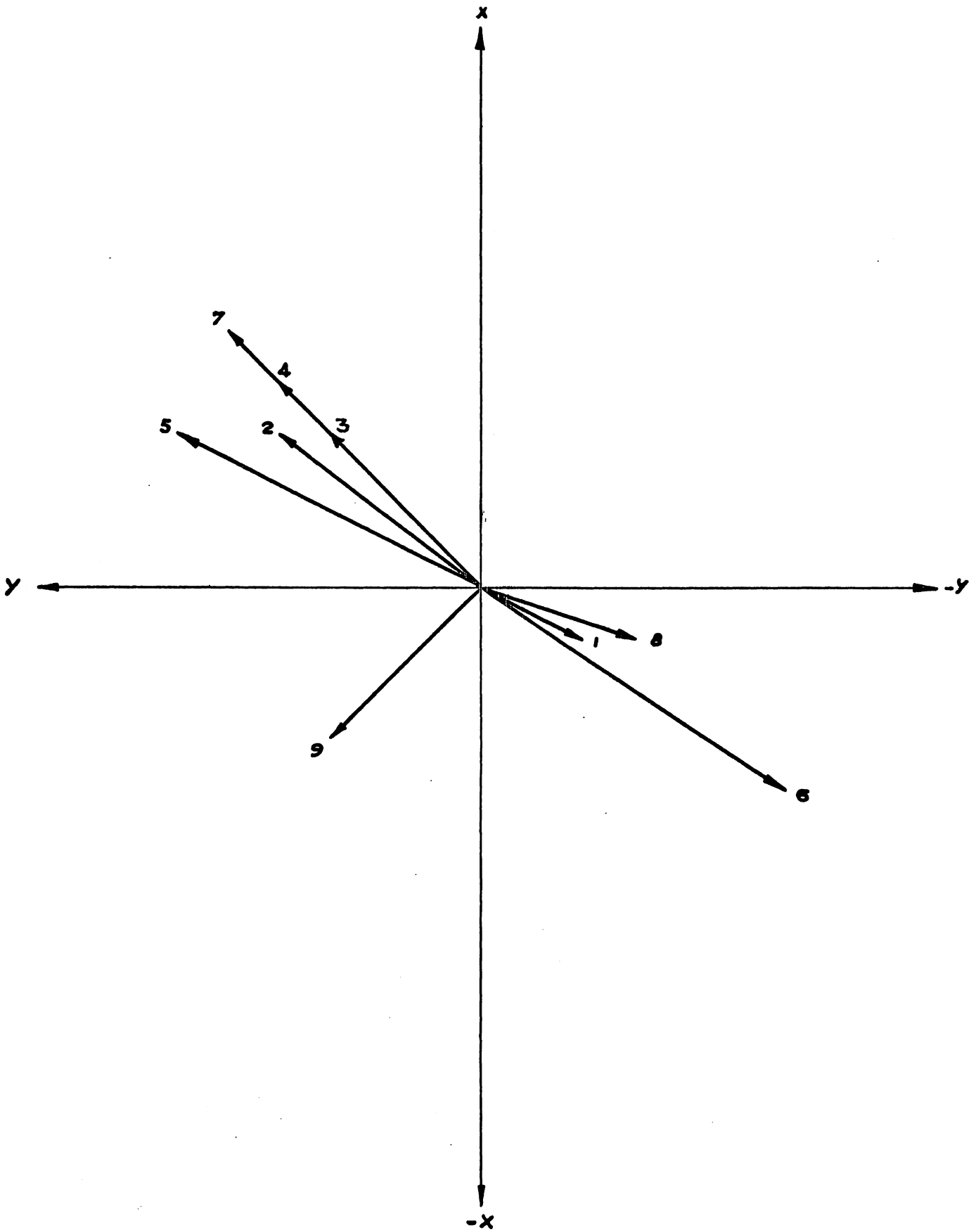


Fig. 19. The derived ellipse from the raw vector diagram Station 4, Urim area.



20. The raw telluric vector diagram, base station. Urim area.

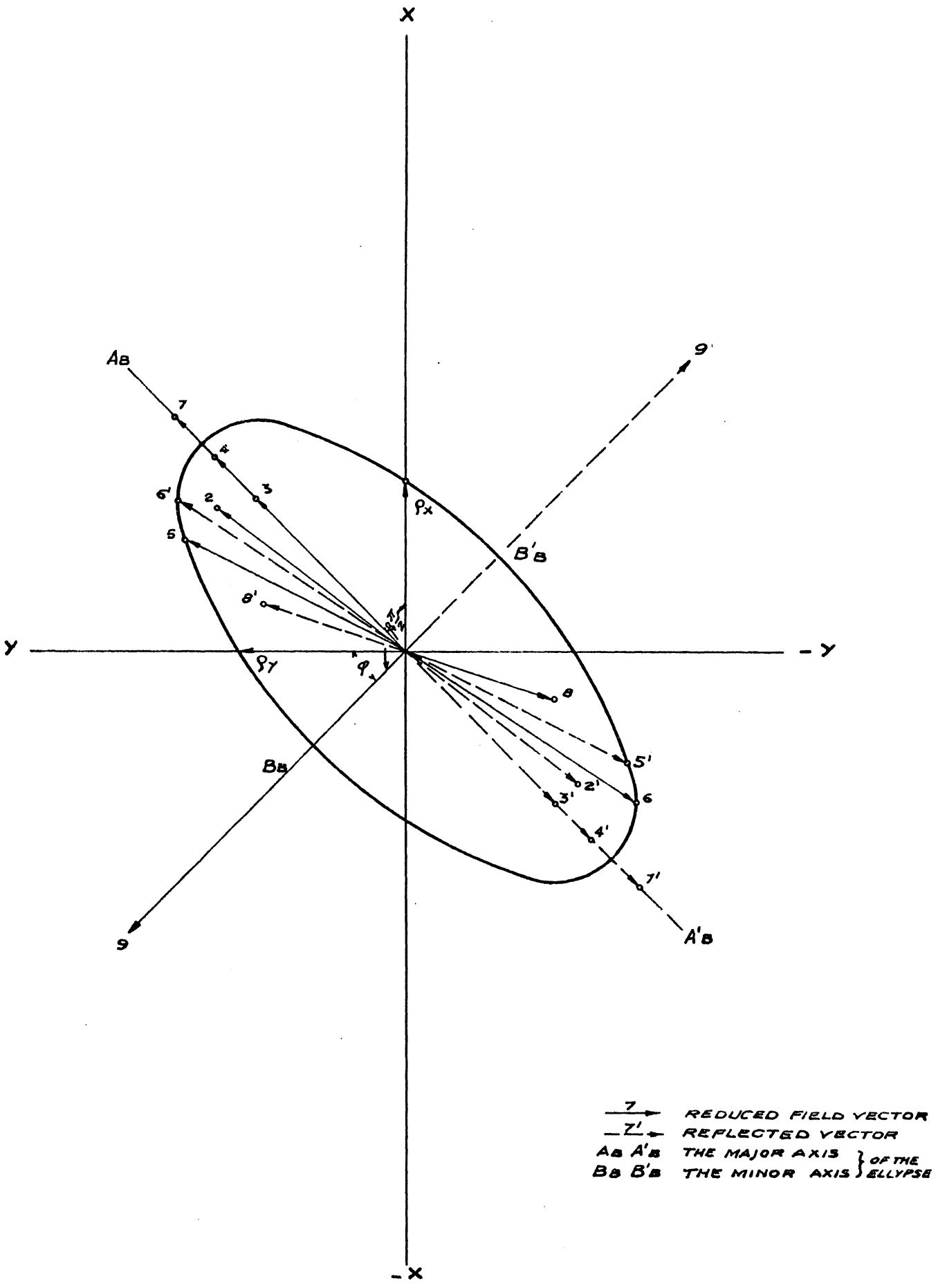


Fig. 21. The derived ellipse from the raw vector diagram, base station. Urim area.

1.1.2 Determination of parameter μ , the amplitudes ratio.

In the case of a linearly polarized telluric current field, the tellurograms showing oscillations without phase shift, the interpretation is done by the method of amplitude ratios.

In interpreting tellurograms in this manner the following steps are taken :

- Synchronization of the tellurograms
- Construction of a vector diagram
- Determination of the parameter and the direction of polarization of the telluric current field.

The synchronization of the tellurograms is done in the same manner as previously described in the ellipse method. (see page 27)

In constructing the vector diagram, the values $\triangle E_{Bx}$, $\triangle E_{By}$, $\triangle E_{Su}$ and $\triangle E_{Sv}$ are taken as the maximum amplitude of a pulse.

These values are plotted along the axes X, Y, and U, V, of a pair of graphs whose azimuths correspond to the actual directions of the corresponding measuring lines and one computes the vectors of the telluric current field $\triangle E_B$ and $\triangle E_S$.

Fig. 22 and 23 show vector diagrams obtained from field measurements, at a base and a field station.

We calculate the value of the parameter μ for each pair of vectors from the expression :

$$\mu_i = \frac{E_{Si}}{E_{Bi}} \qquad (17)$$

The average value of the parameter $\bar{\mu}$ is then calculated :

$$\bar{\mu} = \frac{1}{n} \sum_{i}^{n} \mu_i \qquad (18)$$

Where n is the number of measurements.

For a higher precision one can calculate root-mean-square errors :

$$\sigma = \sqrt{\frac{\sum_{1}^{n} \epsilon^2}{n-1}} \qquad (19)$$

Where : $\epsilon = \mu_i - \bar{\mu}$

The final value for μ is calculated from the equation :

$$\mu = \bar{\mu} \pm \frac{\sigma}{\sqrt{n}} \qquad (20)$$

The direction of the axis of polarization is determined by taking the average azimuth of the vectors :

$$A_{ZB} = \frac{1}{n} \sum_{1}^{n} A_{ZBi} \qquad (21)$$

where A_{ZBi} is the azimuth of the base vector i and

$$A_{ZB} = \frac{1}{n} \sum_{1}^{n} A_{ZSi} \qquad (22)$$

where A_{ZSi} is the azimuth of the station vector i .

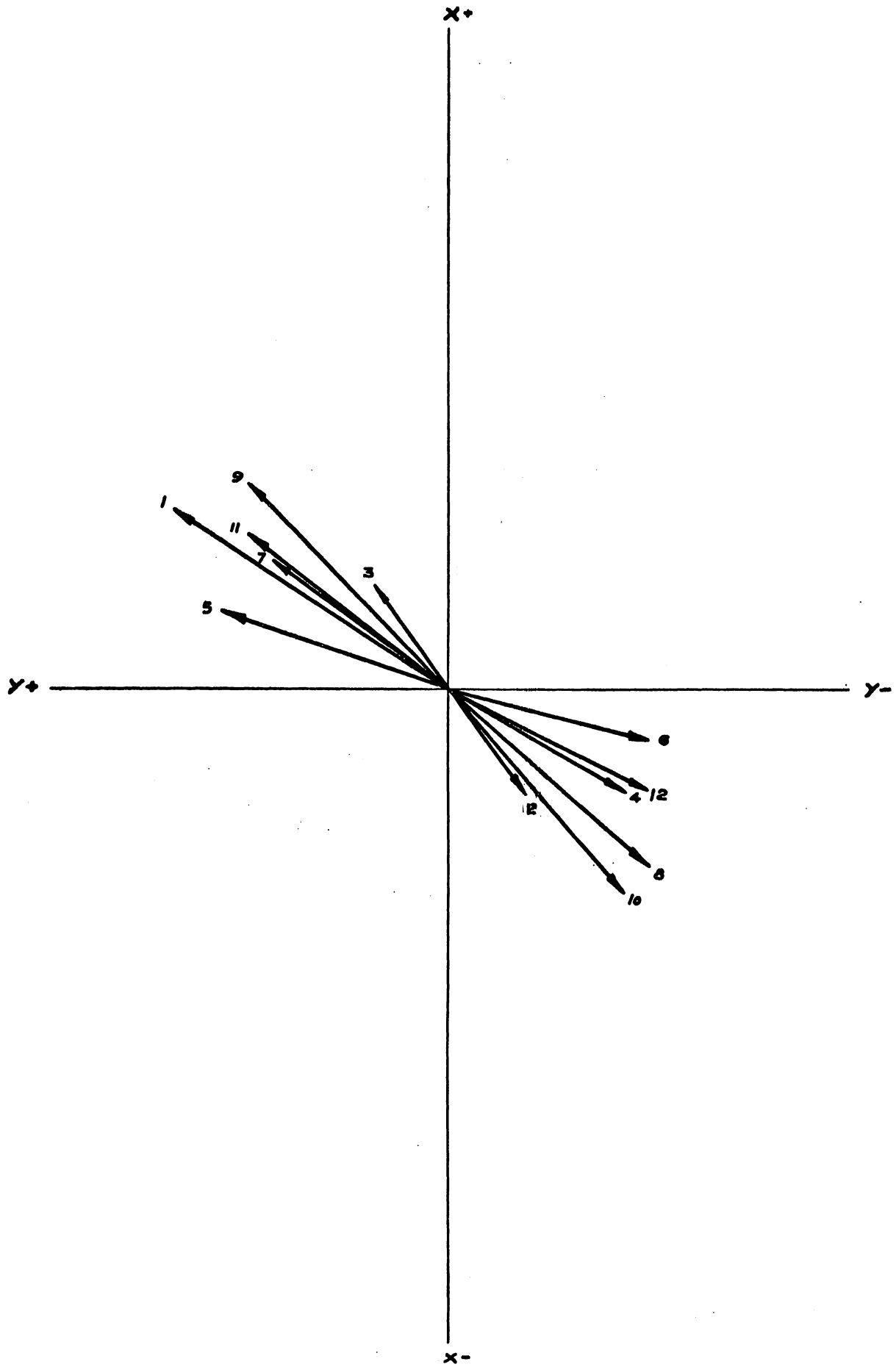


Fig. 22. Vector diagram of a linearly polarized field; base station. Urim area.

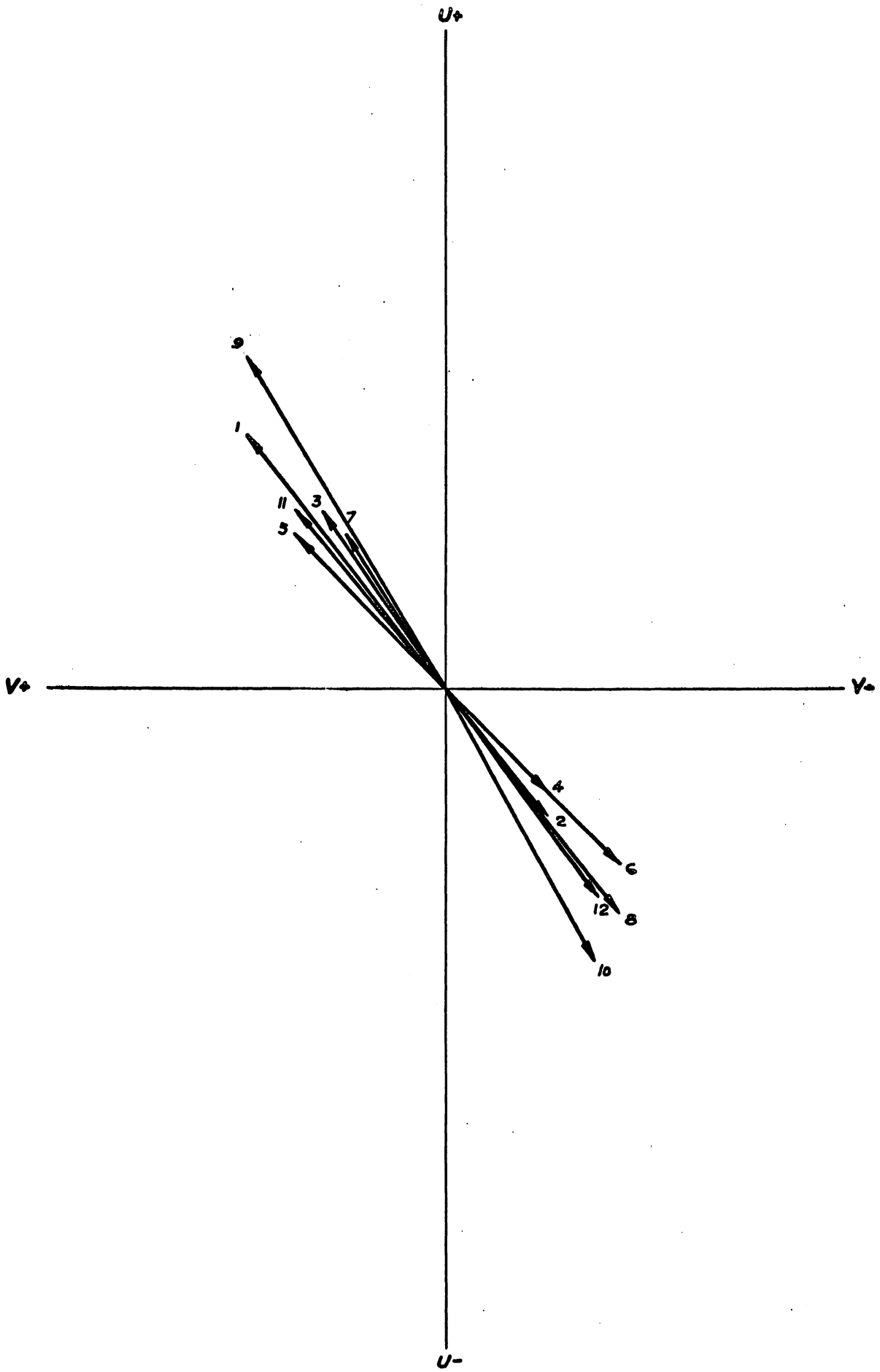


Fig. 23. Vector diagram of a linearly polarized field; field station. Urim area.

1.2 PRESENTATION AND INTERPRETATION OF THE RESULTS OF TELLURIC MEASUREMENTS

1.2.1 Qualitative Interpretation

The observational results of a telluric current survey may be presented in the form of maps (or profiles) which characterise the pattern of variation of telluric currents in the explored area.

1.2.1.1 Contour map of the mean intensity of telluric current.

By using the values of the parameter K, determined for all the stations and taking arbitrarily the average telluric field intensity at the base as 100 standard units, one can calculate the values of the average telluric field intensity at the stations, with the equation:

$$E_{Si} = 100 \times K_i \qquad (22)$$

Where K_i is the value of the parameter for the station i

E_{Si} is the value of the average field intensity at the station i

A map of the average telluric field intensity gives a unique presentation of the behavior of the telluric current field in a given area.

Considering the properties of the geoelectric section as being uniform, the contour map exhibits the main topographic features of the electrical basement surface. Maximum of the values of average field intensity correspond to elevations of the electrical basement and minima to depressions. From this, a general idea of the form of basement structure is provided.

In the same way one can also obtain a profile of the mean intensity of telluric currents which indicates the form of the basement along this profile.

1.2.1.2 Contour map of the parameter M

In preparing this map, the values of parameter M obtained from the field data are plotted on the base map of the surveyed area. The parameter M at the base station is arbitrarily assumed to be unity. Contours of the values of parameter M are drawn.

In the geological interpretation of this map, one identifies the zones with significant changes of the value of the parameters. Such variations usually accompany abrupt changes of the total longitudinal conductance of the sedimentary column, and indicate large-scale structures. The flanks of an anticlinal structure may be indicated by maximum or minimum values for the parameter M, depending on the location of the base station. The location of a steep monoclinal fold in the electrical basement can also be indicated.

1.2.1.3 Map showing field ellipses

This map is obtained by plotting on the base map the field ellipses in a reduced scale.

In interpreting such a map one considers the regularities in the change of orientation of the field ellipses, having in mind that, in areas with monoclinal dip of the electrical basement or broad anticlinal structure, the principal axes of the ellipses are oriented along the dip and the strike directions of the electrical basement.

1.2.1.4 Map showing field intensity vectors

This map is a representation of the vectors E_s constructed for the field stations, corresponding to the initial vector E_B at the base. Current lines may also be shown on the map by sketching them parallel to the vectors, E_S. By this, one obtains a graphic presentation of the telluric current flow, the current flow patterns being related indirectly to structure, providing the properties of the geoelectric section are consistent.

1.2.1.5 Contour map for the parameter μ

In areas with stable linear polarization of the telluric current field, such that the direction of the polarization axis remains constant within 10-15°, it is possible to prepare maps for the parameter μ . The map is obtained by plotting values of the parameter μ , on a base map and a value of unity is assigned for μ at the base station.

This map is analogus to the contour map of the average field intensity and the geological interpretation is based on the same principles. Maximum values of the parameter μ correspond to elevations of the electrical basement and mini-

mum values of μ correspond to depressions, providing that the electrical properties of the geoelectric section are consistent. The structure of the electrical basement is better reflected when the telluric field is polarised across the strike of the structure and not along the strike.

1.2.2 Quantitative interpretation

Quantitative interpretation of the telluric current surveys, which consists of determination of the structure contours on the surface of the electrical basement, is accomplished using statistical methods or simplified analytical methods of interpretation.

1.2.2.1 Statistical methods of interpretation

It is possible to use empirically established correlations between average field intensity and the properties of the sedimentary section known from electrical sounding, seismic prospecting, or drilling at a few isolated points in the area.

The correlation between the average field intensity E, and the total longitudinal conductance S of the sedimentary column is studied. The values of S are taken from reference electrical soundings and the relationship between E and S is plotted on bilogarithmic paper, providing a conversion of the values of E to values of S for the area.

The values of the average longitudinal resistivity ρ_s for the sedimentary column are determined from electrical soundings, made at boreholes or at points where the depth of the electrical basement is known from seismic work.

The thickness of the sedimentary column above the electrical basement is calculated with the formula :

$$H = \rho_s \times S \qquad (23)$$

1.2.2.2 Analytical methods of interpretation

These methods may be used for simple elongate structures when the electrical basement coincides with the surface of high resistivity crystalline basement and the sedimentary column has uniform resistivities, without containing highly resistant screening beds.

The calculations are performed using theoretical formulas derived from the two-dimensional problems.

A traverse is made across the center of the structure at right angles to the strike. At a point along the profile, where the thickness H_o of the sedimentary column is known from geologic, electric or seismic data the field intensity vector E_o, directed across the structure is established as reference. An arbitrary value is taken for this vector. Vectors E_q, field intensity vectors corresponding to this reference vector E_o, are calculated for the remaining points, along the profile by using the coeficients a, b, c, d. Considering that $_s$, average longitudinal resistivity of the sedimentary column does not change substantially and that the slopes of the surface of the electrical basement are reduced (10 - 15°) one can calculate the approximate thickness of the sedimentary column H_q using the formula :

$$H_q = H_o \cdot \frac{E_o}{E_q} \qquad (24)$$

In general for an <u>electrical basement of high resistivity</u> which is covered by a sedimentary column with uniform relatively low resistivities, one can use the approximate expression.

$$\frac{E_S}{E_B} \simeq \frac{H_B}{H_S} \qquad (25)$$

or :

$$\frac{E_S}{E_B} = K \qquad (26)$$

so :

$$K = \frac{H_B}{H_S} \qquad (27)$$

in which :

E_B = field intensity vector at the base

E_S = field intensity vector at the station

H_B = depth of electrical basement at the base

H_S = depth of electrical basement at the station.

By using this relation for all the measured points in the area one can obtain an approximate image of basement structure.

As an example we present the profile Negba-Kokhav which was calculated by this method (using equation 27) and assuming an arbitrary value of the depth at the electrical (resistant) basement (fig. 10).

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